

Fleet Sizing and Coordinated Trajectory Planning for Continuous Age-of-Information Data Collection with UAV Swarms

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Abstract—A fleet of K unmanned aerial vehicles (UAVs) operates continuously from a central depot to collect data from M ground sensors, minimising the time-average priority-weighted age of information (AoI) under a fixed fleet-level energy inventory—a fixed total modular battery capacity divided among however many UAVs are flown. How many UAVs should be airborne? We derive a two-regime fleet-sizing rule with no fitted constant: a *reach* cap set by the farthest sensor, which we show duty cycling cannot evade but only relocates to the airborne count; and a *commute* cap set by the realised commute radius, whose governing principle—maximise the fleet’s visit rate, proportional to $C/\bar{c}$ —follows exactly from a cost-weighted square-root allocation of visit frequencies under regular visits, a separable long-run marginal-cost model, and a K -invariant normalised marginal-cost profile, and whose closed form is a first-order approximation that we show underpredicts in an identified high-budget regime. A closed-form boundary, $r_{\max}/r_c = 1 + \sqrt{P/P_f}$, predicts which cap binds, and an analytical pre-deployment criterion, computable from the same inputs, flags fields in which the reach cap may not govern within the operating window. On 240 simulated deployments across three layout families, four sensor counts and three budgets, its closed-form point predictor matches the swept optimum to within one UAV in 78% of instances, and estimating its commute inputs before deployment, from the site survey and known system parameters without a pilot flight or fleet sweep, leaves all 272 point predictions and regime classifications unchanged; where the criterion passes, the median AoI regret of using the rule instead of a sweep is zero; in the clustered (paper) family the 90th percentile is below 5%, and no passing instance in any family loses more than 27%. Fleet coordination through shared route commitments proves essential: without it, 35–71% of visits land on sensors another UAV has just served and the objective is $2.6\times$ worse. The optimum is more stable across planners than the objective is: an age-blind patrol lies within one UAV of our planner’s optimum in 90% of 216 paired deployments although their AoI differs by up to 23% depending on the layout, and in a controlled comparison with a CP-SAT planner that proves 96.6% of its decisions optimal, all three planners lie within one UAV of each other and of the point predictor in every instance. Under a published k -means clustering policy, whose AoI is 32% worse than ours on clustered fields and 31% better on the core layout, the optimum lies within one UAV of the point predictor in 84% of deployments, and a territorial scheduler we build from that comparison lowers the published policy’s AoI in every layout family (median 7–21%, in 204 of 216 held-out deployments) and ours on the core and ring layouts, while its optimum lies within one UAV of the point predictor in 88%. Explicit event-state information is worth $\leq 1\%$ at the optimum within the event model studied and is harmful for short-lived events; we bound that one channel of advantage rather than claiming learned policies cannot help.

Index Terms—Age of information, UAV data collection, fleet sizing, multi-UAV coordination, Internet of Things, trajectory planning.

I. INTRODUCTION

UNMANNED aerial vehicles are increasingly used as data mules for sensor networks deployed where fixed backhaul is impractical: environmental and agricultural monitoring, infrastructure inspection, and disaster response [1], [2]. In such deployments the relevant performance measure is not throughput but freshness: how old the most recent data from each sensor is when the operator needs it. Age of information (AoI) captures this directly [3], and priority weights capture that some sensors matter more than others. We study *age at collection*: a sensor’s age resets when a UAV downloads its buffer, not when the UAV lands and delivers to the sink. This collection-side freshness is the quantity this paper optimises, because it is the quantity a data mule controls; we do not claim it is the standard definition of AoI. It differs from receiver-side AoI, which also includes the return leg to the sink, and we state it here so that no reader infers the latter.

A large literature optimises the trajectory of a single UAV, or of a fixed number of UAVs, for AoI [4]–[6]. The question addressed here is prior to trajectory design: given a field, an airframe, and a total energy budget, *how many UAVs should the fleet have?* The budget here is a shared inventory — a fixed total battery capacity divided among however many UAVs are flown (Remark 1) — not a battery per airframe; with independent per-airframe batteries and no other shared bottleneck the question has no finite answer, and this paper does not address that case. Splitting a fixed budget across more UAVs buys parallelism but gives each less range, so the answer is not monotone. We show that it is governed by two physically distinct mechanisms, that each has a closed form, and that a simple boundary predicts which one binds for a given field.

The problem is formulated for continuous operation: UAVs launch, fly a sortie, return, and relaunch indefinitely, and the objective is the long-run time-average of priority-weighted age. This differs from the finite-mission formulations common in the literature and changes the mathematics in ways we make explicit. In particular, sensors that no UAV can reach do not merely add cost; their cost grows linearly with the operating horizon and diverges as it lengthens, so over long horizons the reach limit acts as a constraint rather than a trade-off, and over finite ones abandoning a sensor can pay.

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The contributions are as follows.

- **A two-regime fleet-sizing rule** (Result 1): a prescription that returns the closed-form *point predictor* $\ell = \min(\lfloor K_{\text{reach}} \rfloor, \text{round } K_{\text{commute}})$ where a design-time test says the reach cap governs within the operating window, and an interval where it does not. We distinguish its three components by evidential status throughout: the reach cap is an *exact feasibility bracket*, between a full-buffer cap that guarantees service from any state and a flight-only cap beyond which service is impossible, 2.5–7% apart here; the commute cap is an *analytical approximation* under a stated visit-rate model; and the combination, together with the finite-window condition, is an *empirically validated principle*. No component contains a fitted constant.
- **Two supporting observations**: under drop-head overflow the informational age of a sensor equals the time since its last visit, so buffer capacity cancels from the objective (Proposition 2); and the empirical “efficiency factor” $\eta \approx 0.9$ used by single-branch reach laws is the expected maximum sensor radius of the placement model, which we give in closed form (Proposition 5).
- **Fleet coordination as state**. Sharing route commitments through the depot removes duplicate targeting, reduces the objective $2.6\times$ at the optimum, and shifts the optimum by one UAV. We quantify the failure mode of the uncoordinated fleet.
- **A coordinated rolling-horizon planner**, attaining 97.8–98.4% of the exact per-decision optimum on average on instances small enough to solve exactly ($M \leq 15$), and 97.7% on proven-optimal decisions at $M = 40$, with an ablation isolating what each component contributes.
- **Robustness of the optimum to the planner**. Under two age-blind patrol policies and a CP-SAT planner that proves most of its per-decision problems optimal, the measured fleet-size optimum stays within one UAV of our planner’s in 90–100% of instances while the objective differs by up to 23%; the rule predicts the patrol’s optimum as well as it predicts ours. Under a published clustering policy that beats ours on one layout and loses on another, the rule predicts the optimum within one UAV in 84% of deployments, more often than for our own planner. We show why the patrol can win on some geometries and that better per-decision solutions do not monotonically improve the long-run objective. A territorial scheduler we build from that comparison beats the published policy in every layout family and our annealer on the core and ring layouts, though not on clustered fields under tight energy, and the rule predicts its optimum at least as well as the published policy’s.
- **The value of information**. At the rule’s optimum, a rolling-horizon age-index planner is within 1% of a belief-augmented planner for long-lived events and strictly better for short-lived ones. Explicit event-state information provides at most about 1% improvement under the tested event model and information structure, which bounds one channel of possible advantage, not

every channel.

Section III states the model. Section V derives the rule. Section VI describes the planner and the coordination mechanism. Section VII validates the rule and its exceptions on 2,891 simulation runs across 272 deployments and reports the information-value experiment. Section VIII discusses limitations, and Section IX concludes.

II. RELATED WORK

Which freshness metric. Age of information is conventionally a receiver-side quantity: the age of the newest update at the monitor [3]. We study *age at collection*, the time since a UAV last drained a sensor, which Proposition 2 makes exactly $t - t^{\text{last}}$. This coincides with AoI wherever the collecting UAV is itself the consumer of the updates, as in multi-UAV data collection that measures AoI at the UAVs [7] and DRL formulations that minimise the weighted AoI of processes at the UAV [8]. It is also the age, or idleness, of the persistent-surveillance and patrolling literature, where a cell’s age is the time since it was last observed [9]. It differs from sink-side formulations whose AoI also includes flight and offloading time after collection [6]; results stated for one metric should not be read as claims about the other.

AoI scheduling. For a single server with unreliable channels, randomized, Max-Weight and Whittle-index scheduling policies come with performance guarantees, and the latter two are near-optimal in simulation [10], and the Whittle-index policy is exactly optimal in special cases and close to optimal in general [11]. Our planner is an energy-constrained set version of such an index, and Section VII shows why this suffices.

UAV data collection for AoI. UAV-assisted data collection is surveyed in [1], [2], and AoI minimisation in UAV-aided collection is reviewed systematically in [4]. Trajectory design has been treated by joint optimisation of trajectory and resource allocation for a UAV relay [5], by DRL [8], and for multiple UAVs with peak- and average-AoI objectives [6]; these take the fleet as given.

Fleet size. Choosing how many vehicles to operate, jointly with their routes, is classical in operations research: the fleet size and mix vehicle routing problem trades fixed vehicle costs against routing costs [12], and fleet composition and heterogeneous routing are surveyed in [13], [14]. There the vehicles are independent, each with its own capacity, so adding one never shortens another’s route; the trade-off is cost against service, not the division of a shared resource. Prior work has also treated UAV count in several freshness-related formulations. Multi-UAV AoI studies have observed an interior optimum in the number of UAVs, caused by interference among independently powered UAVs [7]; others select the minimum number of UAVs for an AoI objective by mixed-integer programming [15], or minimise the number of UAVs needed to meet battery and data-age constraints [16]. In persistent monitoring, the minimum number of robots meeting revisit-latency constraints with recharging has been bounded by approximation algorithms [17], and single-UAV patrols have been optimised for time-average AoI [18]. These treat fleet size through numerical optimisation, through feasibility

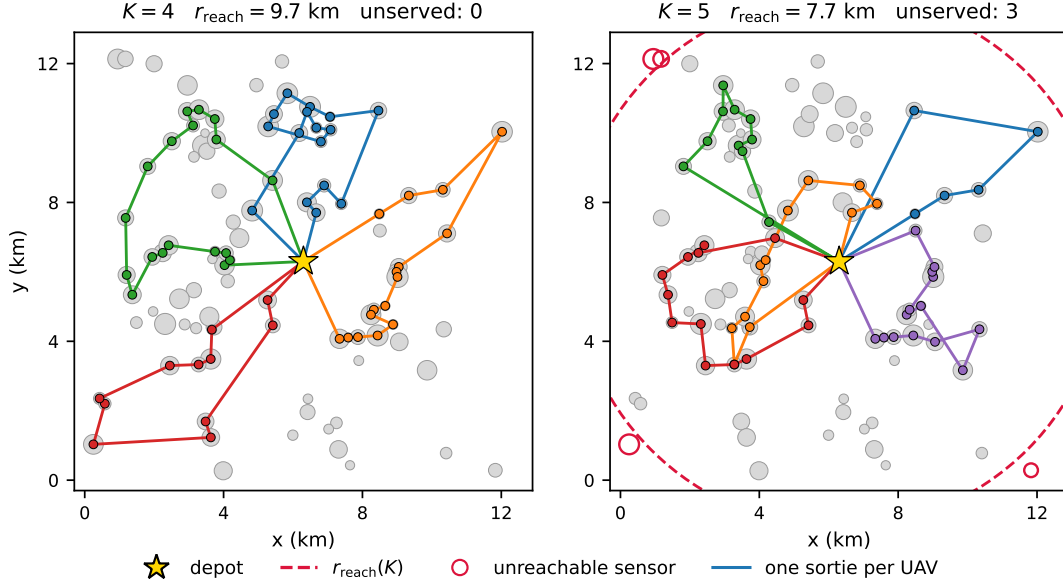


Fig. 1: The reach mechanism. One sortie per UAV on the same instance ($M = 100$, $E_{\max} = 1.5$ MJ, hour 6), at the fleet size given by the rule ($K = 4$) and one UAV beyond it ($K = 5$). Marker area scales with sensor priority; the dashed circle is $r_{\text{reach}}(K)$. Splitting the same battery across five UAVs shrinks the reachable set inside the field: three corner sensors (red) fall outside it and are never served, so their age grows for the rest of the deployment. The served-set cost also rises, from $J = 6.9 \times 10^5$ to 1.1×10^6 .

constraints on independently powered agents, or both; with independent batteries, adding an agent never shortens another's range. We have not found work that derives a design-time rule for the AoI-optimal airborne fleet size under a fixed fleet-level energy inventory, in which each additional UAV adds parallelism and reduces the energy of every sortie, or that identifies the reach and commute regimes studied here. The interference optimum of [7] is the case Remark 1 anticipates: a finite optimum under independent batteries requires a different shared bottleneck, there spectrum.

Learned policies. Learned combinatorial optimisation is surveyed in [19], and attention-based policies have been trained for routing problems [20]. Section VII-L bounds one channel through which a learned policy could gain here — explicit event-state information — not every channel, and Section VII-K tests the fleet-size optimum under independent planners.

III. SYSTEM MODEL

A. Field, fleet and energy

M sensors sit at fixed positions x_i in a square field of side L with a depot at the centre; $r_i = \|x_i - x_0\|$ is the distance of sensor i from the depot and $r_{\max} = \max_i r_i$. Sensor i has baseline priority $w_i > 0$ and generates data at rate λ_i into a buffer of capacity B bits with drop-head overflow (oldest data discarded). A UAV serves a sensor by hovering over it and uploading its buffer at rate R_u , taken constant: the UAV hovers directly above the sensor at fixed altitude, so the air-to-ground link is at its shortest range and near-certain line of sight for the whole of the upload. A range- or channel-dependent rate would replace the fixed R_u by an achievable rate at the hover point and therefore alter only the dwell time

TABLE I: Notation.

M, K	sensors, UAVs
L	field side; depot at the centre
$r_i, r_{\max}$	sensor radius from depot; largest of them
r_c	realised commute radius (depot to first/last stop)
$E_{\max}, \rho, U$	energy inventory, reserve fraction, usable $U = (1 - \rho)E_{\max}$
E_u	per-sortie budget U/K
$v, P_f, P_h, \bar{P}$	speed, flight/hover/mean productive power
R_u, B, λ_i	uplink rate, buffer size, generation rate
w_i, g	sensor priority; event-age coefficient
$\tilde{A}_i$	informational age of sensor i (Prop. 2)
t_c, n, T_s, τ_s	commute time, stops, sortie time, time per stop
$K_{\text{reach}}, K_{\text{flight}}$	full-buffer and flight-only reach caps
K_{commute}	commute cap
$\mathcal{P}(t)$	fleet route commitments with ETAs
$b_i, q, \bar{\tau}$	event belief, event correlation, event lifetime

$t_d = \min(\lambda_i \tilde{A}_i, B)/R_u$; both caps take R_u only through the cap-dwell term $P_h B/R_u$ in (3) and through τ_s in (6), so the structure of the rule is unchanged and its constants shift. We do not model inter-UAV interference: each UAV's uplink is assumed independent of the others'. The coordination rule of Sec. III-D prevents two UAVs from serving the same sensor but does not deconflict trajectories or radio links; uploads are short relative to sortie duration, which limits simultaneous transmission, but we do not claim that it removes interference. Sensor-side transmit energy and duty cycling are likewise outside the model; they would enter as an upper constraint on t_d , in the same place as the buffer cap, and therefore act on both branches exactly as B/R_u does.

The fleet has K UAVs sharing a usable energy inventory $U = (1 - \rho)E_{\max}$, where ρ is a safety reserve, so each sortie has budget $E_u = U/K$.

Remark 1 (What the shared budget means, and when it applies). Every result below follows from $E_u = U/K$, so the assumption deserves to be stated plainly. We are not modelling K independent airframes each carrying its own battery; there, per-sortie energy is a constant of the airframe, total energy is proportional to K , and no reach cap exists. We model the case in which the energy the fleet can carry aloft is fixed and adding airframes divides it: K simultaneously airborne UAVs share an inventory U , so each carries U/K . This is the situation of a fixed total modular battery capacity divided equally among the simultaneously airborne UAVs — a configurable number of battery modules per airframe, with swapping and recharging fast enough not to bind — or of fleets compared at equal total battery capacity — equal battery mass or battery cost — rather than equal per-airframe capability. Concretely: a field team deploys with a fixed pool of battery modules and airframes that accept a variable number of them; some modules fly while the rest recharge. Flying two airframes with four modules each or eight with one each is exactly the choice of K at fixed U . Shared battery pools serving several UAVs through automated swap-and-recharge stations are established practice for persistent missions [21], [22]; what the model adds is the modular bay, which lets the airborne share of the pool be divided among however many airframes fly. Without it, a swap station gives every sortie a whole pack whatever K is, which is the independent-battery case below. U is a stock, not a rate: $E_{\max}$ is the total energy inventory, not a single battery, and each UAV's sortie is limited by its share of it. It is not a model of a charging station of fixed power. With all K UAVs airborne and instant turnaround, the fleet draws about $K\bar{P}$ on average (0.3–0.4 kW per airborne UAV here), which grows with K ; a charger of fixed power P_c therefore imposes a different constraint, $K\bar{P} \leq P_c$, a cap on the airborne count that does not act through E_u and that we do not model. Where energy is not the shared resource—each airframe carrying its own fixed battery $E_u = E_0$, with abundant charging and no other bottleneck—neither cap yields a finite optimum. The reach radius no longer shrinks with K , so (3) does not bind; and since n and T_s are then K -independent, the fleet visit rate $\Gamma \approx Kn_0/T_{s,0}$ is monotone increasing, so the commute branch does not bind either. Commute overhead still makes each sortie less efficient, but it does not make an additional independently powered UAV harmful. The non-monotone fleet-sizing problem studied here therefore arises from a shared fleet-level resource; with independent per-airframe energy a finite optimum would require some other shared bottleneck — charging throughput, spectrum, operator attention, or airspace congestion near the depot — which we do not model.

An operator who owns more airframes than the inventory usefully supports could try to evade $E_u = U/K$ by flying them in rotation: keep each airborne a fraction δ of the time, so that the inventory is shared among $K\delta$ airborne UAVs and each sortie carries $U/(K\delta)$. Proposition 1 shows this does not evade the cap; it relocates it to the number of UAVs simultaneously airborne, which is the quantity the rule sizes. The assumption in Section III that all K UAVs are airborne at all times is therefore the normalisation $\delta = 1$, not an independent restriction.

Battery mass. Dividing U among K airframes also divides its mass: each carries $E_u = U/K$ of battery, so a smaller fleet flies heavier; and rotor power rises with take-off mass (hover power as $m^{3/2}$ by momentum theory). Because K is fixed within a deployment, each airframe's mass is constant during it, and every per- K result holds with P_f and P_h evaluated at that K 's load; the reach cap becomes the fixed point $K = vU/(2P_f(U/K)r_{\max} + vP_h(U/K)B/R_u)$. This does not carry over to the closed form K_{commute} , which differentiates $\Gamma(K)$ at constant power; with mass-dependent power the commute cap is the numerical maximiser of Γ . The constants we use are therefore the powers at one reference load. To size the effect, take a 2.0 kg airframe, 180 Wh/kg packs, our P_f, P_h at a 0.5 kg pack, hover power $\propto m^{3/2}$ and flight power $\propto m^\gamma$ with $\gamma \in [0.5, 1.5]$ (the exponent depends on how much of flight power is induced). At the reach cap the battery is only 12–17% of take-off mass, and the fixed point, unique in every instance, lies 3–10% above the constant-power K_{reach} (medians over the 240 primary instances for $\gamma = 0.5$ and 1.5), changing $\lfloor K_{\text{reach}} \rfloor$ in 58 to 203 of them. That is the same size of error as the input misstatements of Section VII-J: the rule is exact only if P_f and P_h are the powers of the airframe carrying the reach-cap load, and specifying them at another load costs one to two UAVs. Simulating mass-dependent power ($\gamma = 1$; $M = 100$, all three layouts, both budgets, 72 instances) confirms the direction and the remedy. The measured optimum rises, from a mean of 7.8 to 9.1 UAVs, moving in 67% of instances. The constant-power rule then falls to exact agreement 0.18 and within-one 0.68 (median regret 10.3%); the fixed-point rule recovers 0.44 and 0.79 (4.3%), and 0.75–0.83 exact on the clustered cells, where the reach cap binds. It does not help on the ring, consistent with the closed-form commute cap not carrying over; our numerical commute cap scales $\bar{P}$ with flight power, an approximation we have not refined.

Proposition 1 (Duty cycling does not evade the reach cap). *Let the energy inventory U be shared among the UAVs airborne at any time, and let each of K airframes be airborne a fraction $\delta \in (0, 1]$ of the time, resting on the pad otherwise, so that a sortie carries $E_u = U/(K\delta)$. Write $K_{\text{air}} = K\delta$ for the expected number of UAVs airborne. Then*

- (i) *every sensor can be served from any state iff $K_{\text{air}} \leq K_{\text{reach}}$, and some sensor is unreachable iff $K_{\text{air}} > K_{\text{flight}}$, with both caps exactly as in Section V-A;*
- (ii) *the fleet visit rate is $\Gamma = K_{\text{air}}n/T_s$ with n evaluated at $E_u = U/K_{\text{air}}$, i.e. (6) with K replaced by K_{air} ;*
- (iii) *consequently reachability and the aggregate visit-rate model depend on (K, δ) only through K_{air} ; under stationary, well-staggered duty cycling with K -invariant visit-interval statistics the same holds for the long-run objective J , and the rule of Result 1 sizes K_{air} .*

Proof. Substituting $E_u = U/(K\delta)$ into (2) gives $r_{\text{reach}} = v(U/K_{\text{air}} - P_h B/R_u)/(2P_f)$, and requiring $r_{\max} \leq r_{\text{reach}}$ gives the first half of (i); the same substitution in $K_{\text{flight}} = vU/(2P_f r_{\max})$ gives the second. A UAV completes a sortie every T_s/δ on average and serves n sensors per sortie, so the fleet rate is $K\delta n/T_s$, giving (ii). The guaranteed and impos-

sible sets, and the aggregate visit-rate model, then depend on (K, δ) only through K_{air} ; service of the intermediate gray set additionally depends on the scheduling policy. The served-set cost additionally depends on the inter-visit variability $(1 + \text{CV}_i^2)$ and, with events, on when UAVs are airborne rather than only how many; the stated staggering condition is what makes these invariant, giving (iii). The physical conclusion—that duty cycling does not evade the resource trade-off—does not need the stronger exact statement. $\square$

Three qualifications. First, the airframe bounds the share: a sortie cannot carry more than the battery capacity E_{batt} , so $U/K_{\text{air}} \leq E_{\text{batt}}$, i.e. $K_{\text{air}} \geq U/E_{\text{batt}}$. The physical battery therefore enters the model as a *floor* on the airborne fleet, complementary to the reach cap from above. Second, $K\delta$ need not be an integer; the equivalence holds in expectation, with the airborne count fluctuating around it and the coordination protocol seeing a varying fleet. Third, this is a small gain for the rule: since the airborne fleet is continuous, an operator can sit at $K_{\text{air}} = K_{\text{reach}}$ exactly rather than at $\lfloor K_{\text{reach}} \rfloor$.

The practical reading is that buying more airframes and flying fewer of them at once buys nothing and costs nothing for freshness: it is the same fleet in the air. An operator with K airframes should fly $\min(K, K^*)$ of them continuously and leave the remainder as spares. This also disposes of the objection that a fifth UAV should not reduce the energy available to the first four—it does not, if the fifth is left on the ground.

Remark 2 (Equal allocation). *We allocate U equally. This is not assumed optimal: with heterogeneous allocation the operator could fund one long-range sortie for a peripheral sensor and divide the remainder, which removes the abandonment that drives the reach branch. Equal allocation is the simplest symmetric policy for identical airframes drawing on one shared inventory. The gap matters most exactly where our reach branch is weakest: the core family (one peripheral outlier, all other sensors near the depot) is the case in which a single long sortie dominates, and is also the family in which the finite-window condition fails (Sec. VII-C). We regard heterogeneous energy allocation as the natural next question rather than a defect of the present model. Section VII-C tests the simplest case — one sortie reserved for the peripheral sensor — and, in the two cells tested, finds that it removes the stranding regime and restores the reach mechanism for the remaining UAVs.*

A UAV flies at speed v with flight power P_f and hovers with power P_h ; flying distance d costs $P_f d/v$ and hovering for t costs $P_h t$. A sortie starts and ends at the depot, must respect E_u including the return leg, and a UAV relaunched immediately on landing. All K UAVs are airborne at all times except during turnaround, which we take as instantaneous.

B. Events

Some sensors experience *events*: transient periods during which their data is urgent. Events fire as a Poisson process with rate ν_i (ν_{hot} for sensors inside hotspots, ν_{cold} otherwise), last a random duration $\tau \sim \mathcal{U}(\tau_{\text{lo}}, \tau_{\text{hi}})$, and are cleared

by a visit. A UAV observes a sensor's event state only on arrival. In Section VII-L we additionally let events be spatially correlated: a hotspot fires as a unit and each member goes live with probability q .

C. Objective

Let $\tilde{A}_i(t)$ be the age of the oldest data still held at sensor i (Proposition 2 shows this is simply the time since the last visit) and let t_i^f be the fire time of a live, uncaught event at i . The fleet minimises

$$J = \frac{1}{T} \int_{T_b}^{T_b+T} \left[\sum_{i=1}^M w_i \tilde{A}_i(t) + \sum_{i \in \mathcal{E}(t)} g w_i (t - t_i^f) \right] dt, \quad (1)$$

where $\mathcal{E}(t)$ is the set of sensors with a live uncaught event, g prices event age relative to baseline priority, T_b is a burn-in, and T the measurement window. The first term is priority-weighted buffer age; the second is event AoI. We refer to them as J_{age} and J_{event} .

Remark 3. *An earlier formulation multiplied w_i by g while an event was live, charging the entire buffer backlog as urgent the instant an event fired. That rewards visiting stale sensors that happen to have events rather than catching events quickly. Equation (1) charges the event's own age.*

D. Fleet state and coordination

At any time the fleet state is $(\tilde{A}(t), \mathcal{P}(t))$, where $\mathcal{P}(t) = \{(j, \hat{t}_j)\}$ is the set of sensors committed in the routes of currently airborne UAVs with estimated arrival times. A UAV that plans at time t excludes from its candidate set every sensor that appears in the remaining route of any other airborne UAV, regardless of which would arrive first; commitments are reported to the depot at launch and at every replan, the depot recomputes their estimated arrival times from the committing UAVs' current positions, and every planner sees the latest state. No inter-UAV link is assumed beyond the depot. Because a committed sensor is excluded from every other planner's candidate set, commitments are not pre-emptible: the first UAV to claim a sensor keeps it, so no UAV can have its target taken while in transit. Section VII-H shows that allowing UAVs to abandon a leg in response to later state changes does not pay. Section VII-G quantifies what happens without this.

E. Decision structure

A UAV decides its route at launch and, optionally, re-optimises its remaining route after every stop from its current position and remaining energy (*per-leg replanning*). Both variants are evaluated. Any routing policy—optimiser or learned—operates within this structure with the same information, which is what makes the comparison in Section VII-L fair.

IV. BUFFER CANCELLATION

Proposition 2 (Buffer cancellation). *Under drop-head overflow and full drainage at each visit, define the informational age of sensor i as the age of its oldest retained sample plus the time span covered by samples destroyed by overflow since*

the last visit. Then $\tilde{A}_i(t) = t - t_i^{\text{last}}$ for any generation-rate process, and the buffer capacity B enters the objective (1) only through the dwell-time cap $t_d \leq B/R_u$.

Proof. After a full drain at t_i^{last} the buffer refills. Before saturation the oldest retained sample was generated at t_i^{last} , so its age is $t - t_i^{\text{last}}$ and nothing has been destroyed. After saturation at $t_i^{\text{last}} + \tau'$, the oldest retained sample has age τ' (the buffer holds exactly the most recent τ' seconds of data in a fluid model) and the destroyed samples cover the span $[t_i^{\text{last}}, t - \tau']$ of length $t - \tau' - t_i^{\text{last}}$. The sum is $t - t_i^{\text{last}}$ in both cases. Defining the destroyed charge as a time span rather than a bit count makes the identity independent of $\lambda_i(t)$. The only place B survives is the upload time at a visit, $\min(\lambda_i \tilde{A}_i, B)/R_u$. $\square$

Two consequences matter for what follows. Age is reset at collection, not at delivery to the depot; this is the “age at collection” convention of Section I and is stated as such. And partial drainage would break the identity, so the planner never partially serves a sensor: if the remaining energy cannot cover a full service plus the return leg, the sensor is skipped.

V. THE FLEET-SIZING RULE

A. Reach branch

Any sortie that serves sensor i must fly at least r_i from the depot to reach it and at least r_i back (triangle inequality), and the direct out-and-back sortie attains this minimum, so $2P_f r_i/v + P_h t_{d,i}$ is the least energy with which i can be served. The dwell $t_{d,i} = \min(\lambda_i \tilde{A}_i, B)/R_u$ depends on how long i has waited, which gives two bounds rather than one. A sensor with a *full* buffer can be served iff

$$r_i \leq r_{\text{reach}}(K) = \frac{v(U/K - P_h B/R_u)}{2P_f}, \quad (2)$$

so every sensor can be served from any state iff

$$K \leq K_{\text{reach}} \equiv \frac{vU}{2P_f r_{\text{max}} + vP_h B/R_u}. \quad (3)$$

With no dwell at all the farthest sensor is out of reach iff $K > K_{\text{flight}} \equiv vU/(2P_f r_{\text{max}})$, which is therefore the exact upper limit. Between the two the farthest sensors remain serviceable only while they are revisited often enough that their buffers never fill. The bracket is narrow — $K_{\text{flight}}/K_{\text{reach}} = 1 + vP_h B/(2P_f r_{\text{max}} R_u)$, between 1.025 and 1.072 here — but it contains an integer in 119 of 272 instances, and our planners do sustain such schedules: at paper $M = 50$, 3 MJ, $K = 11$ ($K_{\text{reach}} = 10.77$, $K_{\text{flight}} = 11.28$) the two sensors beyond the full-buffer radius are each visited 44–47 times after burn-in, with their buffers never more than 17% full. We use the full-buffer cap K_{reach} — the *robust* reach cap, which guarantees service from any state without assuming a revisit schedule — as the reach branch of the rule. Substituting K_{flight} raises exact agreement from 0.57 to 0.65 with within-one agreement unchanged (Sec. VII-B), which shows that part of the rule’s small positive bias in reach-bound fields is this bracket rather than any error in the mechanism. We nevertheless keep K_{reach} , because the gain from operating

inside the bracket is bought with a failure mode the steady-state grid cannot show: a gray-set sensor is lost permanently as soon as its buffer fills, so a single service interruption longer than its fill time B/λ_i removes it for the rest of the deployment. Section VII-B quantifies the trade-off and gives the condition under which the flight-only cap is the better choice. For $K > K_{\text{flight}}$ at least one sensor is never served, and for $K_{\text{reach}} < K \leq K_{\text{flight}}$ one is lost as soon as its buffer is allowed to fill. Formally, at fleet size K the sensors split into three sets: $\mathcal{S}_{\text{guar}}(K) = \{i : 2P_f r_i/v + P_h B/R_u \leq U/K\} = \{i : r_i \leq r_{\text{reach}}(K)\}$, serviceable from any buffer state; $\mathcal{A}_{\text{imp}}(K) = \{i : 2P_f r_i/v > U/K\}$, which no schedule can serve; and the schedule-dependent *gray set* $\mathcal{G}(K)$ of the remainder, served only while a planner keeps their buffers from filling. An abandoned sensor’s contribution to (1) is $w_i(T_b + T/2)$, which grows linearly with the window and diverges as $T \rightarrow \infty$; for finite T it is a finite cost that can rationally be traded against better service to the rest of the field, which is what the core family does. Writing $J(K) = J_s(K) + \sum_{i \in \mathcal{A}(K)} C_i(T)$ with J_s the served-set cost and $\mathcal{A}(K)$ the abandoned set, we have:

Lemma 1 (Finite-window condition, one step). $J(\lfloor K_{\text{reach}} \rfloor) < J(\lfloor K_{\text{reach}} \rfloor + 1)$ whenever

$$\sum_{i \in \mathcal{A}(\lfloor K_{\text{reach}} \rfloor + 1)} w_i \left(T_b + \frac{T}{2} \right) > J_s(\lfloor K_{\text{reach}} \rfloor) - J_s(\lfloor K_{\text{reach}} \rfloor + 1), \quad (4)$$

This is a *local* crossing condition: it rules out $\lfloor K_{\text{reach}} \rfloor + 1$, not every larger fleet. A fleet can be worse at one more UAV and better at five more, which is exactly what the core family does. The global statement $K^* = \lfloor K_{\text{reach}} \rfloor$ requires in addition that $J(K) \geq J(\lfloor K_{\text{reach}} \rfloor + 1)$ for all $K > \lfloor K_{\text{reach}} \rfloor + 1$, or a one-crossing property we have not proved. The design-time criterion (5) below is the global form we actually use: it compares the surrogate objective at $\lfloor K_{\text{reach}} \rfloor$ with every larger fleet rather than one step. The left side grows with the window; the right side does not. As stated, (4) requires J_s at two fleet sizes and so is not checkable before running the system, and it compares against only the next fleet size, whereas an operator who abandons a sensor may keep adding UAVs as long as the served set keeps improving. The design-time form therefore evaluates the surrogate objective of Proposition 4 at every larger fleet size instead of one step. With the radial cost surrogate $c_i^{\text{rad}} = 2P_f r_i/v$ and $C(K) = \bar{P} \Phi(K)$, write

$$F(K) = \frac{Q_K^2}{2C(K)} + \left(T_b + \frac{T}{2} \right) \sum_{i \notin \mathcal{S}_{\text{guar}}(K)} w_i, \quad Q_K = \sum_{i \in \mathcal{S}_{\text{guar}}(K)} \sqrt{w_i c_i^{\text{rad}}},$$

the served-set freshness at the ideal allocation plus the age cost of every sensor without a full-buffer guarantee. The criterion is

$$F(K) > F(\lfloor K_{\text{reach}} \rfloor) \quad \text{for every integer } K \in (\lfloor K_{\text{reach}} \rfloor, K_{\text{max}}], \quad (5)$$

where K_{max} is the largest fleet with $\Phi(K) > 0$. It asks whether $\lfloor K_{\text{reach}} \rfloor$ minimises the surrogate over all larger fleets, so it is exact within the surrogate and needs no one-crossing property; the served set is allowed to shrink with K , as it

must. Every quantity follows from the sensor set, the airframe, the commute inputs $(r_c, \bar{P})$ and the operating window, and evaluating it takes a closed-form expression at a handful of integers. Its error relative to the realised objective is the surrogate's, not a bounding error: the served-set term assumes the ideal square-root allocation, which over-states the freshness an extra UAV buys (Sec. VII-C), and the abandoned set is the full-buffer one, which charges gray-set sensors as abandoned although a planner may keep them serviceable and so pushes the verdict toward passing. The latter effect is at most the reach bracket, one UAV in our grid. Sufficient-type variants that pair the smallest abandoned set with the largest served gain over all larger fleets are also valid within the surrogate, but they flag 87–93% of clustered instances in which the rule loses nothing, so we do not use them. The criterion is design-time in the sense that no fleet sweep or simulation is needed, but it takes the commute inputs $(r_c, \bar{P})$, which are measured in this paper; Section VII-J shows that estimating them before deployment changes between 4 and 17 of 164 verdicts, depending on where the estimate is evaluated, and neither of the properties we report. Section VII-D evaluates it on every instance. Section VII-B shows one layout family for which (4) fails at a 12-hour window and tightens at 24 hours.

B. Commute branch

Below the robust reach cap every sensor is serviceable from any state, and the objective is governed by how often sensors are served. Model a sortie as an out-and-back commute of duration $t_c = 2r_c/v$, where r_c is the *realised* commute radius (distance from the depot to the first and last served sensors), followed by n productive stops each of duration τ_s at mean productive power $\bar{P}$, a time-weighted average of hop and hover power (measured 301–400 W here). Then $n = (E_u - P_f t_c)/(\bar{P} \tau_s)$, $T_s = t_c + n \tau_s$, and the fleet visit rate is

$$\Gamma(K) = \frac{Kn}{T_s} = \frac{1}{\tau_s} \cdot \frac{K(U - K P_f t_c)}{U + K t_c (\bar{P} - P_f)}. \quad (6)$$

If served-sensor age is proportional to $1/\Gamma$ (equal weights, no inspection-paradox correction), then $\arg \min_K J_s = \arg \max_K \Gamma$, and setting $d\Gamma/dK = 0$ gives (Appendix A)

$$K_{\text{commute}} = \frac{U}{t_c (P_f + \sqrt{P_f \bar{P}})} = \frac{vU}{2r_c (P_f + \sqrt{P_f \bar{P}})}. \quad (7)$$

Before listing the approximations, one apparent one can be removed outright. The sortie duration

$$T_s = t_c + n \tau_s = t_c + \frac{E_u - P_f t_c}{\bar{P}}$$

does not contain τ_s : the product $n \tau_s$ is fixed by the energy left after commuting, so how that energy is split between many short stops and few long ones changes n and τ_s in compensating ways but not the length of the flight. Writing $\Phi(K) = \tau_s \Gamma(K) = K(U - K P_f t_c)/(U + K t_c (\bar{P} - P_f))$, we have $\Gamma = \Phi/\tau_s$ exactly.

Proposition 3 (Backlog-dependent dwell does not move the optimum). *In steady state with no buffer overflow the fleet*

must collect exactly what the field generates, so the fleet's total hovering time per unit time is $H = \sum_i \lambda_i / R_u$, independent of K . Consequently the dwell component of a stop is $H/\Gamma(K)$, giving $\tau_s = \tau_{\text{hop}} + H/\Gamma(K)$, and therefore

$$\Gamma \tau_s = \Gamma \tau_{\text{hop}} + H = \Phi(K) \implies \Gamma(K) = \frac{\Phi(K) - H}{\tau_{\text{hop}}}.$$

Hence the dwell component cancels exactly, and

$$\arg \max_K \Gamma(K) = \arg \max_K \frac{\Phi(K) - H}{\tau_{\text{hop}}(K)},$$

which reduces to $\arg \max_K \Phi$ if and only if the inter-stop travel time τ_{hop} is itself K -invariant.

This is worth stating because dwell *does* vary with K — larger fleets visit more often, find smaller backlogs and dwell for less time — and one would expect that to matter. It does not: the conservation law makes it cancel identically. The no-overflow condition is what makes the conservation law applicable, and it holds where we use it: on the ring at 3 MJ ($M = 100$, $K = 4, 6, 8$, two seeds) no served visit finds a full buffer and no data are dropped, and at 1.5 MJ the fraction of generated data dropped is 0.02% at the optimum $K = 3$ and at most 0.9% over $K = 2$ –4. Overflow becomes material only once sensors are stranded (71% of generated data lost at $K = 6$, where 68 of 100 sensors are never served), outside the regime in which the proposition is invoked.

The hop component is a different matter, and we do not assume it away. Measuring travel time per stop directly, τ_{hop} is nearly K -invariant in the clustered family ($55.5 \rightarrow 52.6$ s from $K = 2$ to 4, a 5% change) but falls sharply on the ring ($68.2 \rightarrow 29.1$ s from $K = 4$ to 8, a 57% change), because smaller budgets select tighter groups of stops. The ring is precisely where the commute branch binds, so this is not a negligible correction there — and following it through resolves the discrepancy reported in Section VII-C. Substituting the measured $\tau_{\text{hop}}(K)$ into Proposition 3 moves $\arg \max \Gamma$ from 6, where Φ alone peaks, to 8, which is where both the measured fleet visit rate and $\arg \min J$ lie. The closed form's error in the ring family is therefore attributable to a single measurable quantity, with no fitted exponent involved. Nor is the decline a property of our planner: an age-blind route-first patrol shows the same effect on the ring, τ_{hop} falling from 34.3 to 25.1 s between $K = 4$ and 8 against 42.8 to 28.6 s for the annealer (Sec. VII-K). The conservation law also yields a throughput bound in its own right: the fleet pays a fixed hovering power $P_h \sum_i \lambda_i / R_u$ whatever its size, so a deployment whose generation rate exceeds what the energy budget can drain is infeasible at every K .

One approximation remains, and it is load-bearing. First, (7) differentiates through t_c as if it were independent of K , whereas r_c in fact *rises* with K . This follows from the same order-statistic argument as Proposition 5: a sortie's commute radius is set by the *nearest* of the n sensors it selects, so $\mathbb{E}[r_c | n] = \int (1 - F_r(x))^n dx$, which decreases in n ; and n falls with K because each UAV has less energy. Larger fleets therefore fly *shorter* sorties whose first and last stops are *farther* out. Section VII-I measures this directly: r_c rises in 18 of 20 cells, by +17.5% to +71.9% in the clustered family, and is

flat ($\leq 4\%$) on the ring, where the radial distribution is nearly a step function and $\mathbb{E}[\min \text{ of } n]$ barely depends on n . Second, and separately, the step $\arg \min_K J_s = \arg \max_K \Gamma$ needs the priority weights, which (6) does not carry. Proposition 4 supplies them.

Proposition 4 (Optimal frequency allocation). *Let sensor i be visited at frequency f_i at long-run marginal energy c_i per visit, with the fleet's productive energy rate bounded by $\sum_i c_i f_i \leq C$. Under regular visits ($\bar{A}_i$ time-averaging to $1/2f_i$), the weighted age $J_{\text{age}} = \sum_i w_i/(2f_i)$ is minimised by $f_i \propto \sqrt{w_i/c_i}$, giving*

$$J_{\text{age}}^*(C) = \frac{(\sum_i \sqrt{w_i c_i})^2}{2C}. \quad (8)$$

With equal costs this reduces to $f_i \propto \sqrt{w_i}$ and $J^* = (\sum_i \sqrt{w_i})^2/2R$ with $R = \sum_i f_i$.

Proof. Stationarity of $\sum_i w_i/(2f_i) + \lambda(\sum_i c_i f_i - C)$ gives $-w_i/(2f_i^2) + \lambda c_i = 0$, i.e. $f_i = \sqrt{w_i/2\lambda c_i}$; imposing the constraint gives $\sqrt{2\lambda} = \sum_i \sqrt{w_i c_i}/C$, and substituting yields (8). The objective is convex on $f_i > 0$, so the stationary point is the minimum. $\square$

Equation (8) is exactly inversely proportional to C for a fixed set of weights and costs. Across fleet sizes the numerator moves too, since the marginal cost of a visit depends on how sorties are composed. Write $c_i(K) = \bar{c}(K) q_i$, separating a common scale from a normalised profile. Then $\sum_i \sqrt{w_i c_i(K)} = \sqrt{\bar{c}(K)} \sum_i \sqrt{w_i q_i}$, and

$$J^*(K) = \frac{\bar{c}(K) (\sum_i \sqrt{w_i q_i})^2}{2C(K)} = \frac{(\sum_i \sqrt{w_i q_i})^2}{2\Psi(K)}, \quad \Psi(K) \equiv \frac{C(K)}{\bar{c}(K)}. \quad (9)$$

The common scale does *not* cancel. What the invariance of the profile q_i buys is that the bracket is K -free, so

$$\arg \min_K J^* = \arg \max_K \frac{C(K)}{\bar{c}(K)} = \arg \max_K \Psi(K),$$

not $\arg \max_K C(K)$. The quantity $\Psi = C/\bar{c}$, a productive energy rate per unit visit cost, is proportional to, but not in general equal to, the total visit rate; we reserve Γ for the physical visit rate throughout. Substituting the optimal allocation into $\sum_i f_i$ gives

$$R^* = \sum_i f_i^* = \frac{C}{\bar{c}} \cdot \frac{B_S}{A_S}, \quad A_S = \sum_i \sqrt{w_i q_i}, \quad B_S = \sum_i \sqrt{w_i/q_i} \quad (10)$$

so the two coincide only when the costs are equal ($q_i \equiv 1$, whence $A_S = B_S$); for heterogeneous costs they differ by the factor B_S/A_S . That factor is K -free whenever the profile q_i is, so

$$\arg \max_K R^* = \arg \max_K \frac{C(K)}{\bar{c}(K)} = \arg \min_K J^*,$$

and the interpretation stands: minimising weighted age is equivalent to maximising how often sensors are served, not to maximising energy spent productively. Equivalently $J^* = A_S B_S/(2R^*)$. This is the sharper statement, it is what the measurements support — in the ring cells where the closed

form misses, $\arg \max \Gamma_{\text{measured}} = \arg \min J = 8$ while Φ peaks at 6 — and it is why a fleet size that spends more energy usefully can still be worse if each useful visit costs more. What matters here is whether the *profile* of costs across sensors changes with K . Since $c_i^{\text{rad}} = 2P_f r_i/v$ depends only on sensor position, the radial profile shape is K -invariant by construction and what can drift is its scale, through changes in which sensors are served. Measuring this over seeds 1–6, the served set is identical for $K \leq 4$ so the drift is exactly zero, and it grows only once sensors begin to be stranded: 0.017, 0.059, 0.117, 0.197 at $K = 5, 6, 7, 8$. Because the optimum for this cell is $K = 4$, the invariance holds exactly at the operating point, and the $K \geq 5$ values delimit the boundary of validity rather than qualifying the result. What this establishes is invariance of the *radial surrogate* profile; it supports, but does not prove, the corresponding condition for the true route-dependent marginal costs that Proposition 4 requires. Incidentally the served sets are *nested* — raising K only removes sensors, never exchanges them — which is not guaranteed a priori and which we verified as an identity between the drift and the mean-cost ratio. In the constant-per-stop sortie model of (6) the mean cost of a visit is $\bar{c}(K) = \bar{P}(K)\tau_s(K)$, so $\Psi(K) = C(K)/\bar{c}(K) \approx \Gamma(K)$ and the commute branch targets exactly the quantity (9) identifies. The identification relies on that model: where $\bar{P}_{\tau_s}$ itself varies with K — as Proposition 3 shows it does through τ_{hop} — maximising C and maximising $C/\bar{c}$ are different problems, and it is the latter that minimises age.

We do not claim that the planner's index realises this allocation. The square-root rule fixes long-run frequencies; the planner's index $w_j \hat{a}_j/e_j$ is a one-step value-per-joule rule for the set-orienting problem, and equalising it across sensors would give inter-visit times $\propto c_i/w_i$ rather than $\sqrt{c_i/w_i}$. An index that does reproduce the square-root ordering under the energy knapsack is $\hat{a}_j \sqrt{w_j c_j}$ (its value-per-joule ratio is $\hat{a}_j \sqrt{w_j/c_j}$). We tested it: with $c_j = 2P_f r_j/v$ it is 2.1% worse in J than the linear index, worse in 11 of 12 paired instances ($p = .006$, Table III). The two rules answer different questions—steady-state frequency versus myopic sortie value—and the myopic one wins at this horizon, consistent with the index-policy literature on AoI. The cost-weighted allocation is therefore used here to justify the fleet-size $\arg \max$, not the routing index. Two gaps remain. The planner's realised f_i differs from f_i^* , which raises J at every K . If that allocation inefficiency $J_{\text{realised}}(K)/J^*(K)$ is approximately K -independent it changes the level of J without materially moving its argmin; we state this as an assumption rather than a consequence of using one algorithm throughout, and it is testable from exported per-sensor visit frequencies. And regular visits are assumed: for irregular ones the inspection paradox gives $\tilde{A}_i = (1 + \text{CV}_i^2)/(2f_i)$, so (8) is scaled by a variability factor that must be roughly K -independent for the equivalence to survive. We measure it: over seeds 1–6, dropping the left-censored first interval, mean CV of inter-visit intervals is 0.298–0.340 across $K = 2 \dots 8$ with no monotone trend, well below the value 1 a memoryless process would give. The implied inflation $(1 + \text{CV}^2)/2$ is 0.544–

0.558 against 0.5 for perfectly regular service, i.e. 9–12% above the deterministic value rather than the $2\times$ an exponential assumption implies, and with no monotone trend in K it is common to all fleet sizes and does not bias the argmax. Two caveats we do not paper over. Strictly the required quantity is the frequency- and weight-weighted mean of $(1 + \text{CV}_i^2)$, not $(1 + \overline{\text{CV}}^2)/2$, since $\mathbb{E}[\text{CV}_i^2] \neq (\mathbb{E}[\text{CV}_i])^2$; the weighted form is the statistic to report and we give the unweighted one here. And the share of sensors with too few visits to estimate CV_i rises from 0.5% at $K = 2$ to 29% at $K = 8$, so the high- K figures describe the well-served subset. The evidence is therefore good around the operating range and weaker at the top of the sweep; we do not claim the variability question is closed across the whole range. The invariance is also a property of our planner rather than of the problem: under an age-blind patrol the mean CV on the ring ($M = 100$, 3 MJ) falls from 0.33 at $K = 4$ to 0.10 at $K = 8$ (Sec. VII-K). We separate that principle from the *closed form* (7), whose accuracy is bounded by the $r_c(K)$ dependence above and which we therefore present as a first-order approximation. We do not claim it is a bound in general: r_c rises with K in most layouts, which penalises large K and can move the optimum the other way. Section VII-C shows that correcting $r_c(K)$ would not repair the cells where the closed form misses, because r_c varies where K_{commute} does not bind and is flat where it does.

C. The rule and its regime boundary

Result 1 (Fleet-sizing rule). *For a fleet sharing usable energy U from a central depot,*

$$K^* \approx \begin{cases} K_R, & \text{if } K_R < K_0, \\ K_0, & \text{if } K_R \geq K_0 \text{ and (5) holds,} \\ \text{some } K \in [K_0, K_R], & \text{otherwise,} \end{cases} \quad (11)$$

where $K_0 = \lfloor K_{\text{reach}} \rfloor$ with K_{reach} given exactly by (3), $K_R = \arg \max_K R^*(K)$ is the commute optimum, and (7) is the closed-form approximation to K_R . The test in the second branch is the global criterion (5), not the one-step Lemma 1, since a one-step comparison cannot establish a global optimum. In the third case the rule gives an interval, not a point: the objective is nearly flat across it because each abandoned sensor costs less than the served-set gain over the window (Fig. 2, core panel), and no closed form locates the minimum within it. We regard an honest interval as preferable to a point prediction that the data contradict; in our grid the interval $[K_0, K_R]$ contains the measured optimum in 163 of 164 reach-bound instances. It is an empirically validated sizing interval, not an analytical bracket: once selective stranding begins, the served-set cost changes with the served set as well as with the visit rate, so K_R does not bound K^* in theory. Dropping the small dwell term, the reach branch binds if and only if

$$\frac{r_{\text{max}}}{r_c} > 1 + \sqrt{\bar{P}/P_f}. \quad (12)$$

Proof. (11) combines the global criterion (5) with the commute principle of Proposition 4. It does *not* invoke Lemma 1,

which is a one-step comparison and cannot decide between $\lfloor K_{\text{reach}} \rfloor$ and every larger fleet. For (12), $K_{\text{reach}} < K_{\text{commute}}$ with $B/R_u \rightarrow 0$ reduces to $2P_f r_{\text{max}} > 2r_c(P_f + \sqrt{P_f \bar{P}})$. $\square$

Both caps scale as $vU/(P_f \cdot \text{length})$ and differ only in which length. Clustered fields with a few peripheral sensors have r_{max}/r_c well above the boundary and are reach-bound; radially spread fields have $r_{\text{max}} \approx r_c$ and are commute-bound. The boundary is ≈ 2.0 for the airframe parameters of Section VII.

Remark 4 (Why single-branch laws appear to work). *For the placement model of Section VII, $r_{\text{max}}/\bar{r} \in [1.8, 2.4]$ across instances—within 15% of the boundary if the commute radius is taken as the mean sensor radius $\bar{r}$. A reach-only law and a commute-only law then give nearly the same K^* , and no sweep of E_{max} or L can separate them since both caps scale as U/L . The realised r_c is much smaller than $\bar{r}$ in clustered fields (the first and last stops are near-depot hotspot sensors), which is why reach binds there. Separation requires varying the shape of the field, which is what Section VII does.*

D. The geometric constant

If the reach cap is written in terms of the field corner, $K_{\text{reach}}^{\text{corner}} = vU/(\sqrt{2}P_f L)$, an empirical factor $\eta \approx 0.9$ is needed to match measurements. It is not an efficiency of the planner.

Proposition 5 (Geometric constant). $\eta(M) = \mathbb{E}[r_{\text{max}}]/(L/\sqrt{2})$. *Under a placement model in which a fraction u of sensors is uniform on the square and the remainder is confined away from the corners,*

$$\mathbb{E}[r_{\text{max}}] \simeq \frac{L}{\sqrt{2}} - \int_{L/2}^{L/\sqrt{2}} \left(1 - \frac{A_{\text{out}}(x)}{L^2}\right)^{uM} dx, \quad (13)$$

where $A_{\text{out}}(x) = L^2 - \pi x^2 + 4[x^2 \arccos \frac{L}{2x} - \frac{L}{2}\sqrt{x^2 - \frac{L^2}{4}}]$ is the area of the square outside the circle of radius x about its centre. The expression is an approximation: it neglects $P(r_{\text{max}} \leq x)$ for $x < L/2$ and assumes only the uniform sensors can lie beyond the inscribed circle.

Proof. $P(r_{\text{max}} \leq x) = (1 - A_{\text{out}}(x)/L^2)^{uM}$ for $x \geq L/2$ since only uniform sensors can lie outside the inscribed circle; $\mathbb{E}[r_{\text{max}}] = \int_0^{L/\sqrt{2}} P(r_{\text{max}} > x) dx$ and $P(r_{\text{max}} \leq x) \approx 0$ for $x < L/2$ when $uM \gtrsim 15$. $\square$

For $u = 0.3$ and $M = 50, 100, 200, 400$, (13) gives $\eta = 0.851, 0.891, 0.921, 0.944$; Monte Carlo on the full model gives 0.855, 0.893, 0.922, 0.944. The factor is therefore M -dependent, tends to 1, and is unnecessary once the rule is written per instance in terms of r_{max} .

VI. PLANNER AND COORDINATION

The reach branch bounds the fleet's action set whatever the planner: K_{reach} guarantees service from any state and K_{flight} excludes it, for every policy; only whether a planner sustains service inside the narrow bracket between them depends on the policy. The commute branch is not, in general — it depends

Algorithm 1: Coordinated sortie planning (one UAV, one decision point)

Input: state $(\tilde{A}, \mathcal{P})$, position, remaining energy, belief b

Output: route σ

```

1  $\mathcal{F} \leftarrow$  sensors not committed in  $\mathcal{P}$  by another UAV, and
  not already served this sortie
2  $\sigma \leftarrow$  greedy ratio construction over  $\mathcal{F}$ 
3 for  $it = 1$  to  $N_{\text{iter}}$  do
4   with prob.  $p_r$ : draw  $j \in \mathcal{F} \setminus \sigma$  by weight; eject
     served sensors in ascending value-per-joule until  $j$ 
     fits; insert  $j$  at cheapest position
5   otherwise: random insert / delete / 2-opt / swap
6   accept by Metropolis on  $\sum_{j \in \sigma} \tilde{w}_j \hat{a}_j$  if
      $E(\sigma) \leq E_u^{\text{rem}}$ 
7 report  $\sigma$  with ETAs to the depot (updates  $\mathcal{P}$ )

```

on the productive-rate curve $\Gamma(K)$ the policy induces, and (7) compresses that curve into $(r_c, \bar{P})$ only under its constant per-stop approximation, which Proposition 3 shows fails when τ_{hop} varies with K . We therefore need a strong planner both to measure K^* and, ultimately, an independent one to test whether K^* is structural. At each decision point a UAV solves

$$\max_{S \subseteq \mathcal{F}, \sigma} \sum_{j \in S} \tilde{w}_j \hat{a}_j \quad \text{s.t. } E(\sigma) \leq E_u^{\text{rem}}, \quad (14)$$

where $\mathcal{F}$ is the set of feasible, uncommitted sensors, $\hat{a}_j$ the projected age at arrival, $\tilde{w}_j = w_j(1 + (g-1)b_j)$ with b_j the planner's belief that an event is live, and $E(\sigma)$ the energy of visiting S in order σ from the current position back to the depot with hover budgeted at the expected arrival time. This is a set-orienting problem (an orienteering problem whose prize is the weighted age reset and whose budget is energy rather than time); we solve it by simulated annealing with a cost-aware eject-and-insert repair operator (Algorithm 1), seeded by the greedy ratio rule $\tilde{w}_j \hat{a}_j / e_j$. The value function is order-independent; order only affects feasibility.

Belief. The prior-only planner sets $b_j = \nu_j \bar{\tau} / (1 + \nu_j \bar{\tau})$ from known rates. The kernel planner additionally raises b_j for sensors near a recently observed live event (Gaussian in distance, exponential decay over $\bar{\tau}$) and sets $b_j \rightarrow 0$ for sensors just visited. Section VII-L compares them.

Learned λ . A visit reveals λ_i exactly (backlog divided by age when the buffer is uncapped); the planner replaces the nominal rate after the first visit.

Hover budgeting. Hover energy is budgeted at the *expected arrival time*, not at the decision time: the buffer fills during the outbound flight, so a decision-time estimate systematically under-budgets hover and forces the UAV to abandon the tail of its route. Budgeting at arrival, using half the fleet's running-mean sortie duration, reduces truncated sorties from 50% to under 8% and lowers J by 16% at $K = 2$; the effect is K -dependent and therefore biases the measured K^* if left uncorrected.

Algorithm 2: Fleet coordination protocol (depot-mediated)

Data: shared commitment set $\mathcal{P} = \{(j, \hat{t}_j, k)\}$ held at the depot

```

1 On decision by UAV  $k$  at time  $t$ , position  $x$ , energy  $E$ :
2   withdraw  $k$ 's own pending commitments from  $\mathcal{P}$ 
3   foreach  $j$  with  $(j, \hat{t}_j, k') \in \mathcal{P}$ ,  $k' \neq k$  do
4     if exclusion mode (all reported results) then
       exclude  $j$ 
5     else set  $\hat{a}_j = \max(0, \hat{t}_{\text{arr}} - \hat{t}_j)$ , exclude  $j$  if
        $\hat{a}_j = 0$  // ETA-projected ablation, Table III
6    $\sigma \leftarrow$  Algorithm 1 on the surviving candidates from  $(x, E)$ 
7   compute ETAs along  $\sigma$  and insert  $\{(j, \hat{t}_j, k)\}$  into  $\mathcal{P}$ 
8 On arrival of UAV  $k$  at  $j$ : remove  $(j, \cdot, k)$  from  $\mathcal{P}$ ;
  broadcast the observed event state and the revealed  $\lambda_j$ .

```

Algorithm 3: Event-belief update (kernel variant)

Input: observation log $\{(j, t_j^o, \ell_j)\}$, hotspot scale σ_h , mean lifetime $\bar{\tau}$

Output: $b_i(t) = \hat{P}(\text{event live at } i)$

```

1  $b_i \leftarrow \nu_i \bar{\tau} / (1 + \nu_i \bar{\tau})$  // stationary prior
  from known rates
2  $\mathcal{O}^+ \leftarrow \{j : t - t_j^o < \bar{\tau}, \ell_j = \text{live}\}$ 
3 foreach  $i$  do
4    $b_i \leftarrow \max(b_i, q \max_{j \in \mathcal{O}^+} e^{-\|x_i - x_j\|^2 / 2\sigma_h^2} e^{-(t - t_j^o) / \bar{\tau}})$ 
5   foreach  $i$  observed within  $\bar{\tau}$  do
      $b_i \leftarrow \min(b_i, \text{prior}_i(t - t_i^o) / \bar{\tau})$  // a visit
     clears the event

```

A. Optimality of the per-decision solver

The value of the rule does not depend on the planner being optimal, but the measurement of K^* does depend on it being good. We certify (14) against an exact solver: for a candidate set of at most 13 sensors, the minimum-energy route through every subset is computed by dynamic programming over subsets with hover folded into the node cost (Held–Karp), and the optimum is the highest-value subset whose minimum-energy closed route fits the budget. The arrival-time hover estimate is frozen at the decision point and shared by both solvers, so node service costs are constants during the subset recursion and the exact optimum is well defined; both SA and the exact solver therefore optimise the same objective over the same feasible set. Running the full simulator on small fields and solving every decision exactly gives Table II: the annealer attains the exact optimum in 81–82% of decisions and 97.8–98.4% of its value on average, against 59–61% and 95.1–95.3% for the greedy ratio rule that seeds it. The two field sizes corroborate each other rather than the result being an

TABLE II: Per-decision optimality against an exact solver. $K = 2$, $E_{\max} = 0.3$ MJ, six seeds, 39 decisions per seed ($n = 234$ per row). Worst case is over individual decisions.

M	planner	mean value / exact	worst	exactly optimal
14	SA with repair (ours)	0.978	0.680	81%
14	greedy ratio rule	0.953	0.622	61%
15	SA with repair (ours)	0.984	0.621	82%
15	greedy ratio rule	0.951	0.522	59%

artefact of the smallest tractable instance. We report the worst case as well as the mean, and note that it moves the wrong way ($0.68 \rightarrow 0.62$ from $M = 14$ to 15) even as the mean improves. Two caveats bound what this establishes. Held–Karp is $O(2^n n^2)$, so the certificate cannot be extended far: exact solution already costs 0.26–0.65 s per decision at $M = 14$ – 15 , and we do not claim near-optimality at $M = 400$. To go further we solve (14) with a CP-SAT model (circuit constraint over optional nodes, energy knapsack, costs rounded so that every accepted route is feasible), which reproduces Held–Karp exactly on 16 decisions at $M = 12$ and closes decisions with up to 37 candidates. On 18 decisions sampled after burn-in at $M = 40$, $K = 4$, every one solved to proven optimality, the annealer attains 0.977 of the optimum on average and 0.931 at worst, consistent with Table II. Where it falls short it tends to under-pack rather than mis-select: it takes fewer stops than the optimum (7 against 12, 14 against 19) while losing only 2–7% of value, and a tenfold iteration budget removes the gap in most but not all such decisions. And per-decision quality compounds: a 2.5–3.4 percentage-point per-decision gap between SA and greedy corresponds to +16.1% in J system-wide (Table III), though the two are measured in different regimes and the ratio is not a transferable multiplier.

B. Planner ablation

Table III isolates each component at the rule’s fleet size on paired instances. Coordination dominates: without it J is $2.6\times$ worse. Mid-leg diversion, arrival-time hover budgeting, the annealer itself and the cost-aware repair operator follow (+30.3, +22.3, +16.1 and +13.0% when removed or added as indicated), each significant in all 12 instances; without arrival-time budgeting 62% of sorties are truncated. Per-leg replanning and the theory-implied index cost 6.6% and 2.1%. Two components make no detectable difference at 12 instances: learning λ_i from visits, and projecting ages from ETAs instead of excluding committed sensors.

C. Decision cost

A launch decision at 1200 annealing iterations has median cost 0.095, 0.124, 0.149 and 0.182 s at $M = 50, 100, 200, 400$ (ten repetitions each, distinct seeds, Intel i9-14900KS pinned to performance cores, idle machine), against sortie durations of 10–30 minutes. Growth is strongly sub-linear: an $8\times$ increase in M costs $1.9\times$ in decision time, because the annealer runs a fixed iteration count whose per-iteration cost scales with route length n , and n is capped by the per-UAV energy budget rather than by how many sensors exist. Planning cost is therefore

TABLE III: Planner ablation at $K = 4$, paper layout, $M = 100$, $E_{\max} = 1.5$ MJ, 12 paired instances, 12 h. ΔJ is relative to the full planner; positive is worse. Sign test drops ties.

	ΔJ	worse in
full planner (SA + repair, coordinated, learned λ)	—	—
— coordination (age vector only)	+160.5%	12/12
— arrival-time hover budgeting	+22.3%	12/12
— greedy ratio rule instead of SA	+16.1%	12/12
— cost-aware repair operator	+13.0%	12/12
+ mid-leg diversion ($\gamma = 1.25$)	+30.3%	12/12
+ per-leg replanning	+6.6%	12/12
theory-implied index $\hat{a}\sqrt{wc}$	+2.1%	11/12
<i>no detectable difference at 12 instances:</i>		
— learned λ_i (nominal rate)	+1.3%	9/12
ETA-projected ages instead of exclusion	−1.1%	4/12

governed by how far a UAV can fly, not by how much there is to look at. For comparison, exact solution of a decision at $M = 14$ – 15 costs 0.26–0.65 s (Table II) — the heuristic is faster in absolute terms at seven times the field size. Computation may sit at the depot or on the airframe: every input a decision uses is known at the depot, a computed route is tens of bytes and the state a UAV would need instead is under two kilobytes, so the protocol is one short exchange per decision either way rather than continuous telemetry.

VII. EVALUATION

A. Setup

A discrete-event simulator implements Sections III–VI exactly, including asynchronous relaunch, in-flight energy accounting, the reserve rule, drop-head buffers, and events. Parameters: $L = 12.6$ km, $v = 20$ m/s, $P_f = 300$ W, $P_h = 400$ W (hover costs more than forward flight, as for a rotorcraft; the measured productive power $\bar{P}$ lies in [301, 400] W across the grid), $R_u = 2$ Mb/s, $B = 50$ Mb, $\lambda_i \sim 5$ kb/s $\times \mathcal{U}(0.5, 1.5)$, $w_i \sim \mathcal{U}(1, 10)$, $\rho = 0.2$, $g = 5$, $\nu_{\text{hot}} = 1/3600$, $\nu_{\text{cold}} = 1/21600$ s $^{-1}$, $\tau \sim \mathcal{U}(45, 90)$ min, $T_b = 3$ h, $T = 9$ h unless stated. Three layout families: *paper* (15 Gaussian hotspots, $\sigma = 0.06L$, 70% of sensors in hotspots, 30% uniform), *ring* (sensors uniform in radius on $[0.8, 1.0] \times 0.45\sqrt{2}L$), and *core* (sensors within $0.2L$ of the depot plus one peripheral outlier fixing $r_{\max}$). Grid: $M \in \{50, 100, 200, 400\}$, $E_{\max} \in \{1.5, 3, 6\}$ MJ, 12 instances per cell, K swept per instance over $[0.5K_{\text{reach}}, 1.5K_{\text{reach}}]$ and extended upward wherever the optimum sat at the top of that range (core $M = 100$ at 1.5 MJ to $K = 20$; core $M = 200$ to $K = 18$ and 34), and, because a top-edge rule cannot detect a lower minimum beyond a range whose optimum is interior, every sweep that ended on a descending slope was then extended in steps of three until it rose again. The core curve has two minima, one near the reach cap and a far one reached by stranding the peripheral sensor; 36 sweeps needed this (all of core $M = 50$, eleven of core $M = 100$ at 3 MJ, one paper instance), and in core $M = 50$ at 1.5 MJ and core $M = 100$ at 3 MJ the far minimum is the lower one. Every sweep in the primary grid now ends on a rising slope, and no optimum is censored. Every annealer sweep also meets the

stricter rule used for the planner comparison of Section VII-K, under which each sweep is extended until its three largest K values all exceed 1.5 times its minimum; extending to it moved no optimum. Simulated annealing uses 1200 iterations per launch. All runs are seeded and reproducible. The *primary* grid is 20 cells $\times$ 12 instances = 240 instances at $L = 12.6$ km; the field-size sweep of Section VII-E adds 4 cells and 32 instances, giving 24 cells and 272 instances in total, from 2,891 coordinated launch-time runs. Twelve instances were generated for every cell without exception, and on the extended sweeps every instance's predicted K lies inside its swept range, so agreement statistics over the primary grid use all 240; Proposition 6 and the estimator comparison are evaluated over all 272. Ablations, certification and diagnostics are separate runs and are counted where reported.

Statistic. For each instance i we compute $K_{\text{reach}}^{(i)}$ from its farthest sensor, $K_{\text{commute}}^{(i)}$ from its measured $(r_c, \bar{P})$, the closed-form point predictor $\ell_i = \min(\lfloor K_{\text{reach}}^{(i)} \rfloor, \text{round } K_{\text{commute}}^{(i)})$, and $\arg \min_K J$. The point predictor is Result 1 with its interval branch suppressed — what the rule would say if the design-time test (5) were ignored — and so is the stricter test of the two; Section VII-D evaluates the prescription with the test applied. In this section “the rule” refers to ℓ_i . We report the agreement rate $P(\arg \min_i = \ell_i)$ over instances, the rate within ± 1 , and, descriptively, two-sided paired sign tests of $J(\ell_i+1) - J(\ell_i)$ and $J(\ell_i-1) - J(\ell_i)$ across instances. *Choice of estimator.* How the measured optimum is summarised across the instances of a cell is a decision that changes the reported agreement, so we state it and its justification explicitly. We take the optimum *per instance* and report the resulting distribution. The reason is not that it scores best but that it is the only comparison at matching granularity: K_{reach} depends on *that* field's farthest sensor and K_{commute} on *that* field's commute radius, so the rule is an instance-level quantity and must be compared against an instance-level optimum. Summarising J across instances first and taking one argmin per cell compares an instance quantity with a population quantity.

Appendix B reports the three cell-level alternatives; none scores higher, and the four disagree on the summarised optimum in 9 of 24 cells.

B. Validation of the rule

Table IV gives every cell. Figure 2 shows $J(K)$ for one cell of each family; Figure 3 plots the served-set optimum against r_{max}/r_c with the boundary (12).

Overall. Across the 240 primary-grid instances (20 cells) the per-instance rule is exact in 56% (95% interval [51, 60]%) and within one UAV in 78% ([76, 81]%). In 13 of 20 cells every instance is within one. The deviations are concentrated in two families and are not distributed noise: the three 3 MJ *ring* cells (commute closed form under-predicts) and the *core* family (finite-window condition), each analysed below. Excluding the seven cells with within-one agreement below 0.9 — ring at $M = 50$ and 100 (3 MJ), and core at $M = 50$ (1.5 MJ), $M = 100$ (both budgets) and $M = 200$ (both budgets) — the

rule is exact in 74% of instances ([68, 80]%) and within one in every one of them ([98, 100]%). Replacing the full-buffer cap K_{reach} by the flight-only cap K_{flight} of Section V-A raises exact agreement over the 240 instances from 0.56 to 0.63 and leaves within-one agreement at 0.78; the paper-family bias falls from +0.19 to −0.08. That gain assumes uninterrupted operation. To price it, we took six instances in which the choice matters and the flight-only cap wins, ran each at both fleet sizes over a 24 h horizon, and grounded the whole fleet for 3 h mid-run, as weather might. Without the grounding the flight-only fleet size has 5–22% lower AoI in all six. With it, the flight-only fleet size is 12–81% worse in five of the six, and six of its seven gray-set sensors are never served again; at the robust cap every sensor recovers. The losses are the ones buffer fill time predicts: every gray-set sensor with B/λ_i shorter than the grounding was lost, the one survivor had a fill time 1.2 h longer than the grounding, and one sensor with only 0.3 h of margin was lost as well, because it had already gone unvisited for part of a revisit interval when the grounding began. We therefore keep the robust cap as the reach branch of the rule, and state the alternative as a condition an operator can check before deployment: the flight-only cap may be used when the longest service interruption the deployment must tolerate is shorter than $\min_{i \in G} B/\lambda_i$ by at least a normal revisit interval. Two borderline cases bracket that margin between 0.3 and 1.2 h here; they do not fix it.

Reach-bound, clustered fields. In all eight paper cells the optimum lies within one UAV of the rule in every instance, with exact agreement 0.42–1.00 and the lower-side sign test significant in every cell. The 6 MJ cell, where events are frequent relative to revisit intervals, obeys the rule (mean optimum 19.25 against $K_{\text{reach}} = 19.2$).

Commute-bound, spread fields. In the three 1.5 MJ ring cells the served-set cost has an interior minimum at $K = 3$ with zero abandonment while $K_{\text{reach}} \approx 4.9$: the commute mechanism, not reach, sets the optimum, and the boundary (12) classifies these cells correctly ($r_{\text{max}}/r_c \approx 1.2$ against 2.0).

Two localised failure families. The aggregate agreement figure is misleading, because the deviations are not scattered: the thirteen cells outside the two families have mean bias +0.25 (+4.0% relative), exact agreement 0.74 and within-one agreement 1.00; the worst bias among them is +0.92. The remaining seven split into two families with distinct mechanisms, analysed in Section VII-C.

A stated domain of validity. Across the 14 reach-limited cells, relative bias correlates with r_{max}/r_c at 0.92, and the cells fall into two groups with nothing between them: −2% to +6% for the nine cells with $r_{\text{max}}/r_c \leq 6.2$, and +42% to +178% for the five from 7.4 upward (+46% at 7.4, +42% at 7.6, +173% at 12.5, +82% at 13.2, +178% at 16.6). Since r_{max}/r_c is dimensionless, obtainable from a single pilot sortie, and already used to classify regime, it doubles as an empirical *warning indicator*, calibrated on the measured commute radius (Section VII-J shows it does not transfer to an ex-ante estimate): the rule is accurate to within roughly 6% while $r_{\text{max}}/r_c \leq 6.2$ and degrades sharply by 7.4. No cell lies between, so this grid places the warning level somewhere in

TABLE IV: Per-instance point predictor versus measured optimum, all cells. n instances; $\overline{\arg \min}$ and $\bar{\ell}$ are means; “+1” and “−1” give the number of instances in which $J(\ell \pm 1) > J(\ell)$ with sign-test p . Regime by mean $r_{\max}/r_c$ against the boundary (12).

layout	M	$E_{\max}$ (MJ)	n	$\overline{\arg \min}$	$\bar{\ell}$	agree	± 1	+1	p	−1	p	$r_{\max}/r_c$	bnd	regime
paper	50	1.5	12	4.50	4.58	0.92	1.00	12/12	<.001	11/12	.006	4.5	2.0	reach
paper	100	1.5	12	4.25	4.25	1.00	1.00	12/12	<.001	12/12	<.001	6.0	2.0	reach
paper	200	1.5	12	4.25	4.00	0.75	1.00	9/12	.146	12/12	<.001	6.1	2.0	reach
paper	400	1.5	12	4.25	4.00	0.75	1.00	9/12	.146	12/12	<.001	6.2	2.1	reach
paper	50	3.0	12	10.00	9.67	0.67	1.00	8/12	.388	12/12	<.001	3.1	2.0	reach
paper	100	3.0	12	9.25	9.17	0.92	1.00	11/12	.006	12/12	<.001	4.9	2.0	reach
paper	200	3.0	12	8.92	8.83	0.92	1.00	11/12	.006	12/12	<.001	6.1	2.0	reach
paper	100	6.0	12	19.25	18.67	0.42	1.00	5/12	.774	12/12	<.001	3.2	2.0	reach
ring	50	1.5	12	3.33	3.00	0.67	1.00	8/12	.388	12/12	<.001	1.18	2.0	commute
ring	100	1.5	12	3.08	3.00	0.92	1.00	11/12	.006	12/12	<.001	1.19	2.0	commute
ring	200	1.5	12	3.00	3.00	1.00	1.00	12/12	<.001	12/12	<.001	1.20	2.1	commute
ring	50	3.0	12	8.50	6.00	0.00	0.00	0/12	<.001	12/12	<.001	1.18	2.0	commute [†]
ring	100	3.0	12	7.75	6.00	0.00	0.25	0/12	<.001	12/12	<.001	1.18	2.0	commute [†]
ring	200	3.0	12	6.92	6.00	0.08	1.00	1/12	.006	12/12	<.001	1.20	2.0	commute [†]
core	50	1.5	12	6.83	4.67	0.50	0.83	8/12	.388	12/12	<.001	7.4	2.0	reach [‡]
core	100	1.5	12	8.33	4.58	0.58	0.67	9/12	.146	12/12	<.001	13.2	2.0	reach [‡]
core	200	1.5	12	13.67	4.92	0.00	0.00	6/12	1.00	12/12	<.001	16.6	2.0	reach [‡]
core	50	3.0	12	10.50	10.08	0.58	1.00	7/12	.774	12/12	<.001	5.8	2.0	reach [‡]
core	100	3.0	12	14.00	9.83	0.42	0.83	6/12	1.00	12/12	<.001	7.6	2.0	reach [‡]
core	200	3.0	12	28.25	10.33	0.08	0.08	6/12	1.00	12/12	<.001	12.5	2.0	reach [‡]

[†]closed form (7) under-predicts; principle holds (Sec. VII-C). [‡]reach cap not the operative bound within the 12 h window: see the design-time condition.

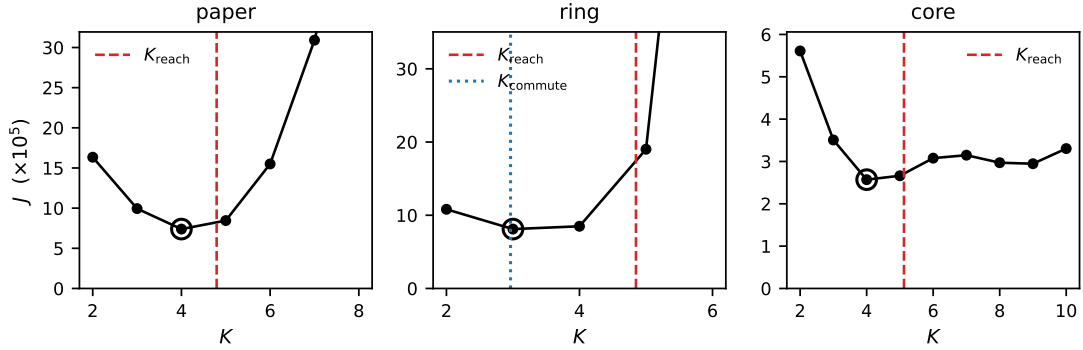


Fig. 2: Mean $J(K)$ with the predicted caps (linear axes; circled marker is the measured optimum), $M = 100$, $E_{\max} = 1.5$ MJ. Paper: reach binds, optimum at $\lfloor K_{\text{reach}} \rfloor = 4$. Ring: commute binds at $K_{\text{commute}} \approx 3$, well below $K_{\text{reach}} = 4.9$, with no abandonment at the optimum. Core: the reach cap is 5, but one abandoned sensor costs less than the served-set gain at a 12 h window, so the curve stays flat past it (condition (5) and Proposition 6).

(6.2, 7.4) without fixing it. We deliberately avoid the word boundary, which this paper reserves for the analytical regime boundary (12); the level is an empirical observation of this study, not an analytical or fitted decision boundary. Five cells lie beyond it and all five degrade by a large margin.

C. The two failure families

Both are now understood, and one of them yields a stronger form of the rule.

The reach cap bounds coverage, not optimality. For $M = 200$ in the core layout the rule gives 4.92 while the measured optimum is 13.67. Exactly one sensor of 200 is stranded from $K = 7$ onward, and J keeps falling well past that point

(6.69×10^5 at $K = 4$ to 3.18×10^5 at $K = 10$). Stranding one sensor in 200 buys nearly three times the fleet. At 3 MJ the effect is larger still: the optimum is 28.25 against a rule of 10.33. This is a real trade-off, not a modelling error or an artefact of our planner — an age-blind patrol also places the optimum far above the reach cap, at 10.75 and 27.67 for the two budgets (Sec. VII-K) — and Proposition 6 prices it. The precise location of the optimum once stranding pays is, however, planner-dependent: the two planners agree to within one UAV at 3 MJ but differ by three at 1.5 MJ.

Equal sharing is what makes this regime hard. The core family is where a single long sortie matters most (Remark 2), so we tested one departure from equal allocation. One UAV

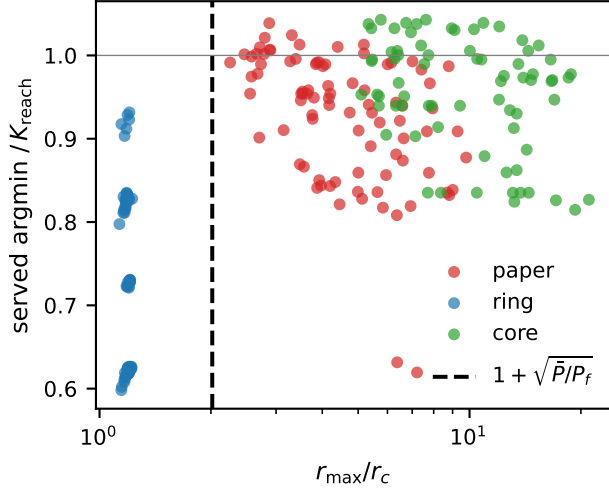


Fig. 3: Served-set optimum normalised by K_{reach} against r_{max}/r_c for every instance, with the boundary (12). Instances left of the boundary optimise below K_{reach} (commute-bound); instances right of it optimise at K_{reach} (reach-bound).

receives just enough energy to serve the peripheral sensor from any buffer state, $1.02(2P_f r_{\text{max}}/v + P_h B/R_u)$, and the other $K-1$ share the rest of U equally; the inventory is unchanged, only its split. On core $M=100$ and $M=200$ at 1.5 MJ, 12 instances each, no sensor is stranded at the optimum in any of the 24 instances, and the AoI at the optimum falls by a median 52% and 20% relative to equal shares. The optimum also becomes predictable again: it lies within one UAV of the reach cap of the $K-1$ equally funded UAVs, which the second-farthest sensor sets, in 23 of the 24 instances and exactly in 15. The reach mechanism therefore appears to apply to whatever part of the fleet shares energy equally: the results suggest funding the outlier's sortie separately and sizing the remaining, equally funded fleet by the rule. We report this as an observation rather than a rule — the split-fleet predictor was formed for this test and evaluated on two cells — and general heterogeneous allocation remains open.

Proposition 6 (Stranding condition). *Let $\mathcal{S}(K) = \mathcal{S}_{\text{guar}}(K) = \{i : r_i \leq r_{\text{reach}}(K)\}$ be the full-buffer surrogate served set and $Q_K = \sum_{i \in \mathcal{S}(K)} \sqrt{w_i c_i}$. Using Proposition 4 for the served set and charging each stranded sensor its age integral over the window, adding one UAV at fleet size K lowers the surrogate objective $J_{\text{age}}^*(C(K)) + \sum_{i \in \mathcal{A}(K)} w_i(T_b + T/2)$ if and only if*

$$\underbrace{\frac{Q_K^2}{2C(K)} - \frac{Q_{K+1}^2}{2C(K+1)}}_{\text{served-set freshness gained}} > \underbrace{\left(T_b + \frac{T}{2}\right) \sum_{i \in \mathcal{S}(K) \setminus \mathcal{S}(K+1)} w_i}_{\text{age cost of newly stranded sensors}}, \quad (15)$$

The sequential stopping point $\hat{K}$ is the smallest K at which (15) first fails.

Three qualifications are deliberate. The condition inherits the full-buffer surrogate: sensors in the gray set $\mathcal{G}(K)$

are charged as stranded although a planner may keep them serviceable, so “stranded” here means “without a full-buffer guarantee”. The condition governs the surrogate objective of Proposition 4, not the planner’s realised J ; Section VII-C measures the gap. And $\hat{K}$ coincides with the global optimum only if the surrogate increments cross zero at most once in K . We have not proved that one-crossing property, so we report $\hat{K}$ as the sequential stopping point predicted by (15) rather than as K^* outright. Equation (15) is the one-step counterpart of the global surrogate test (5): $F(K) - F(K+1)$ is exactly the difference of its two sides, and the global test checks every larger fleet rather than stopping at the first failure. Every quantity in it is computable before deployment: the sensor positions give $\mathcal{S}(K)$ and the radial surrogate $c_i^{\text{rad}} = 2P_f r_i/v$ that we substitute for the true marginal cost, the weights are known, and $C(K) = \bar{P} \Phi(K)$ is closed-form. As elsewhere, using c_i^{rad} is an approximation: the ring analysis shows the true marginal cost moves with route composition, which sensor positions alone cannot capture.

Evaluated per instance on all 272 instances (24 cells), it does what it was derived to do and exposes its own limitation in the process (Table V). On the cell where the plain rule fails worst — core, $M=200$, 1.5 MJ, where it predicts 4.92 against a measured optimum of 13.67 — (15) predicts 14.00, moving exact agreement from 0.00 to 0.67 and within-one agreement from 0.00 to 1.00; at 3 MJ it predicts 22.17 against 28.25 where the plain rule gives 10.33. The mechanism is therefore confirmed rather than merely plausible: the optimum exceeds the coverage bound precisely because one sensor’s age integral is cheaper than the freshness bought for the other 199.

As a general-purpose point predictor, however, it is worse than the plain rule. Across all cells exact agreement falls from 0.57 to 0.44 and within-one agreement from 0.81 to 0.78: (15) removes the catastrophic tail in the core family but overshoots in the clustered family, increasingly with M ($M=200$: 6.00 against a measured 4.25; $M=400$: 7.00 against 4.25, no instance within one). The reason is visible in the derivation. The gain term is evaluated with Proposition 4’s *optimal* square-root allocation, which the planner approximates but does not attain, so the freshness purchased by an extra UAV is over-estimated; and since Q_K^2 grows as M^2 while the newly stranded weight does not, the over-estimate grows with M . An earlier version of (15) held Q fixed at Q_K on both sides; that is not the surrogate difference, and it happened to offset part of this over-estimate (exact 0.51, within-one 0.87). We report the correct condition and its weaker numbers. We therefore report both: $\min(\lfloor K_{\text{reach}} \rfloor, K_{\text{commute}})$ as the point predictor, and (15) as the condition that says when the coverage bound may safely be exceeded and by how much. A tighter gain term — the realised rather than ideal allocation — is the obvious next refinement.

Ring at higher budget: the closed form, not the principle. At 3 MJ the ring optimum lies at 7–8.5 against $K_{\text{commute}} = 6$, with no sensor stranded through $K=9$, so coverage plays no part. Measuring each factor separately localises the error precisely.

The *principle* is exact here. The measured fleet visit rate $1/T_{\text{rev}}$ peaks at $K=8$ — and $\arg \min_K J$ is also 8.

TABLE V: Coverage rule $\min(\lfloor K_{\text{reach}} \rfloor, K_{\text{commute}})$ against the stranding condition (15), selected cells and overall (272 instances, 24 cells).

cell	measured	plain	(15)	plain ± 1
core $M=200$, 1.5 MJ	13.67	4.92	14.00	0.00
core $M=200$, 3 MJ	28.25	10.33	22.17	0.08
core $M=100$, 1.5 MJ	8.33	4.58	8.75	0.67
core $M=100$, 3 MJ	14.00	9.83	11.42	0.83
paper $M=100$, 1.5 MJ	4.25	4.25	5.00	1.00
paper $M=400$, 1.5 MJ	4.25	4.00	7.00	1.00
ring $M=100$, 1.5 MJ	3.08	3.00	3.00	1.00
all cells: exact		0.57	0.44	
all cells: within one				0.81

TABLE VI: $J(24\text{ h})/J(12\text{ h})$, $M = 100$, $E_{\text{max}} = 1.5\text{ MJ}$, six paired instances.

K	2	3	4	5	6
paper ($\lfloor K_{\text{reach}} \rfloor = 4$)	1.01	1.00	1.01	1.23	1.48
ring ($K_{\text{commute}} \approx 3$, $K_{\text{reach}} = 4.9$)	1.02	1.00	1.01	1.18	1.74

Proposition 4's equivalence therefore holds on the nose in the very cell where the closed form misses by two UAVs. What fails is $\Phi(K)$, the model of that rate, which peaks at $K = 6$.

Three candidate causes are eliminated by measurement. The *dwell* component of service time is ruled out by Proposition 3, which shows it cancels identically; the hop component $\tau_{\text{hop}}(K)$ is precisely what does not cancel and is treated below. The productive power $\bar{P}$ is flat at 309–312 W for $K \leq 9$, moving only once the fleet starts stranding sensors. And r_c is flat within $\pm 4\%$ on the ring by construction. What remains is the per-stop energy: the model's stop count $n = (E_u - P_{ftc})/(\bar{P}\tau_s)$ under-predicts the realised count by a factor that *grows* with K (26.4 against 17.0 at $K = 4$, 10.6 against 4.9 at $K = 8$, a ratio rising from 1.55 to 2.16 and beyond). Smaller budgets buy tighter clusters of stops whose intervening hops are shorter, so the marginal energy of a stop falls with K while (6) holds it fixed. This motivates a sublinear-yield diagnostic, but the per-sortie decomposition of Appendix B shows the fitted exponent cannot be read as a hop-energy exponent; the operative correction is the K -dependent τ_{hop} of Proposition 3. A per-stop cost that decreases in K shifts $\arg \max \Phi$ upward, in exactly the observed direction and of the observed size.

A fitted sublinear stop count $n = c(E_u - P_{ftc})^\alpha$ recovers the missing UAVs, but α is fitted, and Appendix B shows it is an elasticity of the closed loop rather than a geometric constant, so we do not adopt it.

Horizon. Table VI reports $J(24\text{ h})/J(12\text{ h})$. Below the reach cap the ratio is 1.00–1.02 (stationary); above it the ratio is 1.18–1.74 (the abandoned term grows with the window). This is the direct signature of the mechanism in Section V-A.

D. Fleet-sizing regret and the design-time criterion

Agreement counts are a strict test of a discrete recommendation, but an operator cares how much freshness is lost by using the closed-form rule instead of sweeping K . Define the regret

TABLE VII: Fleet-sizing regret $J(K_{\text{rule}})/J(K^*) - 1$, conditional on the design-time criterion (5). Commute-bound instances (ring) are classified by the boundary (12), to which the window test does not apply.

family	test	n	median	90th pct
paper	pass	50	0.0%	4.6%
paper	flagged	42	0.0%	1.0%
core	pass	25	0.0%	21.4%
core	flagged	47	24.4%	126.3%
ring	(commute-bound)	72	2.0%	22.4%
all reach-bound, test passes		75	0.0%	10.2%
all reach-bound, flagged		89	0.0%	84.4%

$\text{Reg} = J(K_{\text{rule}})/J(K^*) - 1$. Two questions then matter: how large the regret is where the rule applies, and whether the operator can tell in advance where it does not. The design-time criterion (5) is meant to answer the second, so we apply it to every instance and report regret conditional on its verdict (Table VII).

Three things follow. First, in the clustered family the rule is an essentially free substitute for an exhaustive sweep: half the instances lose nothing and nine in ten lose under 5%. Second, empirically the test behaves as intended: no instance whose regret exceeds 27% passes, and every instance above 30% is flagged—the 47 core instances it flags have median regret 24%, the 25 it passes have median 0%—while its false alarms are harmless: the 42 paper instances it flags have 90th-percentile regret of 1%. An operator who runs the test and, when it flags, sweeps K instead of trusting the rule loses nothing in the reach-bound families beyond the sweep itself. Third, the commute-bound family's regret is a different quantity: the boundary (12) classifies these instances correctly, and their median 2% and maximum 31% are the price of the constant-cost closed form (7), recoverable by maximising the measured visit rate (Sec. VII-C) and not by a better commute radius: the ex-ante inputs of Sec. VII-J leave these instances' predictions, and therefore their regret, unchanged. We report these separately: a pooled figure would flatter the rule where it works and hide where it does not. The flatness that makes the commute optimum hard to pin down is also what makes missing it cheap: at $M = 40$ a fleet one UAV off the optimum costs a median 2.8% in J on the ring, against 18% in the clustered family, and this holds under every planner we tested (Sec. VII-K).

E. Scaling in field size

Both caps scale as $vU/(P_f \cdot \text{length})$, so $K^* \propto 1/L$ at fixed E_{max} . Table VIII tests this axis directly, which the E_{max} sweep cannot: E_{max} and L enter the rule only through the ratio U/L , so varying E_{max} alone cannot distinguish the rule from any other quantity proportional to U . Table VIII gives the result. The optimum falls monotonically from 7.12 to 3.00 as L goes from 8 to 20 km and the rule tracks it from 7.00 to 2.88, with mean bias +0.25 and every instance within one UAV at every field size. The exact-agreement dips at 10 and 16 km are a rounding artefact rather than instability: the bias there

TABLE VIII: Field-size sweep, paper layout, $M = 100$, $E_{\max} = 1.5$ MJ, 8 instances per L (12 at 12.6 km).

L (km)	8	10	12.6	16	20
measured $\arg \min_K J$	7.12	5.75	4.25	3.75	3.00
rule	7.00	5.25	4.25	3.25	2.88
exact agreement	0.88	0.50	1.00	0.50	0.88
within ± 1	1.00	1.00	1.00	1.00	1.00
$r_{\max}/r_c$	5.26	5.71	6.00	5.73	5.77

TABLE IX: The rule with an off-centre depot. Bias is rule minus measured optimum; regret is $J(\ell)/J(K^*) - 1$.

depot	family	n	exact	± 1	bias	regret median
(0.25 L , 0.25 L)	paper	72	0.88	1.00	-0.12	0.0 / 0.0%
	core	72	0.64	0.90	-0.32	0.0 / 0.0%
	ring	72	0.17	0.94	+0.89	10.0 / 21.3%
(0.10 L , 0.50 L)	paper	24	0.83	1.00	+0.08	0.0 / 1.1%
	ring	24	0.00	0.96	+1.04	20.9 / 50.2%

is exactly +0.50, so the mean rule sits midway between two integers and instances round either way. Scaling the field scales $r_{\max}$ and r_c together, so $r_{\max}/r_c$ stays in 5.26–5.77 and every cell remains reach-limited; this sweep therefore varies the magnitude of the optimum over a $2.5\times$ range without changing regime, which is what makes it a clean test of scaling.

F. Depot position

Every result so far places the depot at the centre of the field. Moving it changes nothing in the rule’s form — $r_{\max}$ and r_c are measured from the depot — but it changes the geometry the rule has to summarise. We repeated the grid for all three families with the depot at (0.25 L , 0.25 L) (216 instances) and for the paper and ring families with it near an edge midpoint, (0.10 L , 0.50 L) (48 instances), with the same protocol and the same anti-censoring extension (Table IX). Sweeps that ended on a descending slope were extended as in Section VII-A (13 core sweeps at the quarter-point depot); no optimum is censored.

The reach mechanism transfers: on the clustered and core layouts the rule is as accurate as with a central depot or more so, and ex-ante commute inputs change nothing. The ring does not. There the rule is one UAV too high in almost every instance, and the error grows with the offset, from a median regret of 10% at the quarter point to 21% near the edge. The ring is centred on the field, so an off-centre depot sees its near side at a few kilometres and its far side at up to twice that. The far side binds through commute cost — a sortie there spends most of its energy getting there and back — but the rule’s commute radius is a fleet average dominated by cheap near-side sorties, so K_{commute} is overestimated, never binds, and the rule falls back on $\lfloor K_{\text{reach}} \rfloor$, one UAV too many. The rule has no notion of a commute radius that varies across the field, and a central depot never exposed that. The planner comparison transfers: under the same measurement rule the territorial scheduler of Section VII-K lowers the published method’s own-optimum J at the quarter-point depot by a median 13.5% (core, 72 of 72), 20.0% (paper, 72 of 72) and

TABLE X: Mean J ($\times 10^5$), paper, $M = 100$, $E_{\max} = 1.5$ MJ, 12 instances.

K	2	3	4	5	6	7	8
coordinated, launch-time SA	16.3	9.9	7.4	8.5	15.5	30.9	53.9
uncoordinated	22.9	18.0	19.3	24.9	34.9	51.0	69.9
duplicate-visit fraction	.35	.46	.56	.62	.65	.69	.71
coordinated, per-leg SA	16.6	10.2	7.9	9.1	16.0	31.2	54.1

8.2% (ring, 65 of 72), and near the edge by 22.8% (paper, 24 of 24) and 1.9% (ring, 20 of 24), the last a small margin.

It still lies within one UAV of the optimum in 94–96% of these 90 instances, but in the tested off-centre ring deployments the point prediction is consistently biased upward, by about 0.6 UAV.

G. Coordination

Table X compares the coordinated fleet with the same planner given only the age vector (no commitments). Without coordination, 35–71% of visits arrive at a sensor another UAV reset after the visiting UAV planned it (rising with K); at $K = 4$ the median age at arrival is under five minutes. The objective at the optimum is $2.6\times$ worse and, because the waste grows with K , the measured optimum shifts from 4.25 to 3.33. Coordination is therefore not a refinement but a prerequisite for measuring K^* at all. The mechanism requires only that each UAV report its route to the depot at launch.

H. Per-leg replanning and mid-leg diversion

Re-optimising the remaining route after every stop eliminates sortie truncation (from 1–6% of sorties to zero) but raises J by 4–7% at every K when the per-leg re-optimisation is given one twentieth of the launch-time budget (60 versus 1200 iterations). The optimum is unchanged in all 12 instances. The comparison is computationally confounded — per-leg decisions get 60 annealing iterations against 1200 at launch — so the finding is that *under this computational allocation* frequent replanning does not pay, not that replanning is intrinsically valueless. Likewise the diversion result below is for one margin $\gamma = 1.25$ and does not establish that every diversion policy is inferior. What is robust across both is that the rule’s K^* is unchanged.

Mid-leg diversion. Per-leg replanning still commits a UAV to the leg it is currently flying. A natural question is whether a UAV should abandon that leg when the fleet state changes in transit. Note first that the coordination rule already removes the adversarial case: because a committed sensor is withdrawn from every other planner’s candidate set, the first UAV to commit keeps its target and cannot be pre-empted en route. What remains is opportunity cost—another sensor’s age grows, or an event fires there, while the UAV is between stops. We implemented diversion to test it: whenever any UAV updates its commitments, every in-flight UAV re-evaluates from its interpolated current position and abandons the current leg if an alternative target exceeds the committed one by a factor γ (a margin that prevents oscillation between two targets); the energy already spent on the abandoned leg is sunk. With

$\gamma = 1.25$ this makes the objective *worse* by 30.3%, in all 12 paired instances ($p = .0005$, Table III). The flying already done is forfeit, and the gain from switching to a marginally better target rarely repays it. Committing to a decision and honouring it beats continuously revising it, which is a convenient property for a deployment: the coordination protocol needs no in-flight renegotiation channel.

I. How r_c varies with fleet size

Measuring r_c against K across all 20 primary cells confirms the order-statistic prediction of Section V-B: r_c rises with K in 18 cells, is flat in the ring family, and falls in one. In the clustered family the rise is +17.5% to +71.9% over the swept range; on the ring it is within $\pm 4\%$, as expected where all sensors sit at nearly one radius so the expected minimum of n draws barely depends on n .

This has a consequence for what a corrected $r_c(K)$ could buy, and it is not what one would expect. The two families in which the rule misses (Section VII-C) are exactly the families in which such a correction cannot act: on the ring, where K_{commute} binds, r_c is flat, so the correction is near zero; in the clustered and core families, where r_c varies most, K_{commute} is far from binding (14.7 and 35.5 against $K_{\text{reach}} \approx 5$), so the correction changes an inactive term. We therefore state the $r_c(K)$ dependence as a bound on the closed form's accuracy rather than presenting a corrected form, and locate the two failures in their own mechanisms instead.

J. Predicting r_c before deployment

K_{commute} uses the realised commute radius, which we measure from the runs. That is legitimate for validating the mechanism but circular as a design rule. Two routes make it non-circular, and neither needs a fleet sweep. The order-statistic form $\mathbb{E}[r_c | n] = \int (1 - F_r(x))^n dx$ requires only the radial distribution F_r of the deployment, known from the site survey that fixes where sensors go, together with n from the energy budget; solving the pair as a fixed point gives $\hat{r}_c$ from geometry. This identity is exact for n i.i.d. radial draws, which the planner's stops are not — they are selected on age, priority, route cost and commitments — so it is an i.i.d. order-statistic *approximation* whose accuracy is itself to be tested, not a planner-independent property of the route. Alternatively a single pilot sortie at the largest candidate E_u returns r_c directly, at the cost of one flight rather than a K -sweep over twelve; we evaluate the first route, which needs no flight at all.

We compute every commute input from the site survey, the airframe, the sensors' nominal rates and the inventory, with no flight and no fleet sweep. $\hat{r}_c$ is the order statistic above with F_r the empirical radial distribution of the sensors. $\hat{n}$ comes from the energy budget, $n = (E_u - P_f t_c) / (\bar{P} \tau_s)$, with the inter-stop time taken from the mean nearest-neighbour spacing of the sensors. $\hat{P}$ comes from the conservation law of Proposition 3: the fleet hovers $H = \sum_i \lambda_i / R_u$ UAV-equivalents, so the hover share of productive time is $h = H / (K(1 - t_c / \tau_s))$ and $\bar{P} = P_f + (P_h - P_f)h$. The three are solved jointly with the fleet size at the commute fixed point $K = K_{\text{commute}}(\hat{r}_c(K))$,

which needs no list of candidate fleet sizes; this is the pre-deployment estimator. As a check, and to compare with the measured inputs, we also evaluate them at the smallest candidate fleet size $\lceil 0.5K_{\text{reach}} \rceil$, where the measured values used elsewhere in the paper are taken. Over all 272 instances:

- *The point predictor is unchanged.* Ex-ante inputs change it in none of the 272 instances, at either evaluation point, so its agreement (0.58 exact, 0.82 within one) and regret are exactly those reported with measured inputs, and the regime classification by (12) agrees in every instance.
- *The design-time criterion keeps its properties.* At the commute fixed point (5) changes its verdict on 17 of 164 reach-bound instances, 14 of them toward passing; at the smallest candidate fleet size it changes 4, all toward flagging. In both cases no passing instance loses more than 27%, every instance above 30% is still flagged, and the paper-family pass rows keep a 90th-percentile regret below 5%. The interval $[K_0, K_R]$ that Result 1 returns when the test flags a field is more sensitive to the estimate: computed from ex-ante inputs it contains the measured optimum in 145 of 164 reach-bound instances, against 163 with measured inputs.
- *$\bar{P}$ is accurate.* It tracks the measured value with correlation 0.978 and mean error 1.4 W over a measured range of 303–340 W, against 7.5 W for the best constant.
- *$\hat{r}_c$ is accurate only where it matters.* Compared at the smallest candidate fleet size, where the measured value is taken: on the ring it is 0.93–0.96 of the measured value (10th–90th percentile). In clustered and core fields it is 0.46 and 0.42 of it, and 0.61 and 0.45 even with the measured n , so the error lies in the order-statistic identity, not in $\hat{n}$: the planner's first and last stops are not the nearest of n random draws. $\hat{n}$ itself is about twice the measured value, the nearest-neighbour spacing understating real hops, which affects nothing downstream here.

The estimator needs to be accurate only where its input controls the decision, and the reason is structural. There the reach cap binds and K_{commute} is far above it; an underestimated $\hat{r}_c$ only raises K_{commute} further, and it only moves r_{max}/r_c further from the boundary (12). The commute cap binds only where sensors avoid the neighbourhood of the depot, which is also where the nearest of n radii is a good estimate of the commute radius. We have tested this on one commute-bound family, and do not claim it for commute-bound fields in general. The one quantity that does not survive is the warning indicator of Section VII-B: with $\hat{r}_c$ every reach-limited cell exceeds the warning level (the smallest ex-ante ratio is 9.9), and all six core cells read about 26 whatever their bias, so that indicator requires a measured commute radius, from a pilot sortie, and is not a pre-deployment test.

Sensitivity to the known inputs. Estimating the commute inputs is one source of error; the physical inputs the rule takes as known are another. Misstating one of them changes only the prediction, not the system, so we score the rule computed from a perturbed input against the recorded $J(K)$ of every deployment. The rule is insensitive to P_h and R_u , which

enter only through the small hover term: a $\pm 20\%$ error leaves within-one agreement at 0.80. It is sensitive to U , P_f and $r_{\max}$, because K_{reach} scales exactly with $U/(P_f r_{\max})$ and a 10% error moves it by 10%, one or two UAVs in this grid: exact agreement falls to 0.27–0.38 and median regret rises to 8–14%. The cost is asymmetric. Errors that overstate the fleet push it past the reach cap, where sensors are stranded, and have a 90th-percentile regret of 140–147%; errors in the conservative direction cost about 40%. This sensitivity belongs to energy-limited fleet sizing rather than to the rule: a fleet sweep in a simulator given the same energy parameters would err identically. The practical consequence is that the energy per sortie and the farthest sensor distance must be known to within a few percent, and that an operator unsure of them should err toward less energy and longer distance.

K. Robustness of the optimum to the planner

Every comparison so far varies our own planner, which cannot establish that the measured optimum is a property of the problem rather than of that planner. We therefore measure it under three planners that share nothing with the annealer except the simulator, the energy accounting and the coordination protocol — each honours the commitment set of Section III-D, so none gains or loses from duplicate visits.

- *Tour patrol*. One closed tour through the depot and every sensor (nearest-neighbour construction, then 2-opt) is cut into consecutive sorties by the energy budget: a shared pointer advances around the tour and each launching UAV takes the next run of sensors that fits, skipping sensors committed elsewhere or unreachable on their own. Ages and priorities play no part in selection; they enter only through hover budgeting. This is route-first, cluster-second routing, and what an operator would deploy without this work. It is also the energy-constrained form of the shared-tour policy of [23], in which UAVs are spread along one tour through every sensor and circulate on it; their model has no depot or energy budget, so here the tour is cut into sorties that fit it.
- *Sweep patrol*. The same, with the tour replaced by polar angle about the depot.
- *Cluster patrol*. The clustering method of [23]: the sensors are partitioned by k -means into one cluster per UAV, and each UAV patrols a closed tour of its own cluster, cut into energy-feasible sorties as above. As in the original, ages and priorities play no part in selection; unlike the shared-tour patrols it gives every UAV a fixed territory.
- *CP-SAT*. The per-decision problem (14) solved by constraint programming, as in Section VI-A, cold-started and with a deterministic search budget, so that results do not depend on machine load. It proves 96.6% of 40,164 decisions optimal at its default budget.

The optimum moves less than the objective. Table XI gives the result. The tour patrol’s optimum lies within one UAV of the annealer’s in 90% of instances ([88, 93]%), and the sweep patrol’s in 94% of those it covers ([91, 98]%); at $M = 40$ all three planners lie within one UAV of each other and of the point predictor in all 24 instances, which bounds the rate

TABLE XI: Fleet-size optimum under independent planners. Top: primary grid, 216 paired instances ($M \leq 200$, 18 cells), per-instance argmin on the common K range. Bottom: controlled comparison at $M = 40$, $E_{\max} = 1.5$ MJ, paper and ring layouts, 12 instances each. “=” exact agreement, “ ± 1 ” within one UAV. Agreement with the rule is computed directly, not by transitivity.

planner	=SA	± 1 SA	=rule
annealer (SA, ours)	–	–	0.56
tour patrol	0.63	0.90	0.53
sweep patrol*	0.56	0.94	0.50
cluster patrol [23]	0.48	0.72	0.49
territorial scheduler (C4, ours) [†]	–	0.84	0.54
<i>M = 40, 24 instances</i>			
annealer (SA, ours)	–	–	0.62
tour patrol	0.75	1.00	0.67
CP-SAT	0.75	1.00	0.62
cluster patrol	0.58	0.83	0.42

*paper and ring only; SA on the same 144 instances: 0.65 / 0.85 against the rule; CP-SAT: default search budget on paper; threefold budget on ring, where it is controlled.

[†]own optima on the union K grid, uncensored; exact agreement with SA not computed.

below by 86% at 95% confidence. Figure 4 shows the $M = 40$ curves. The rule predicts the patrol’s optimum about as well as it predicts ours (0.54 / 0.80 against 0.56 / 0.77). This is not because the planners perform alike. The tour patrol is 23% worse than the annealer on clustered fields and 10% *better* on the ring, and the sweep patrol is more than twice as costly on clustered fields; CP-SAT improves on the annealer by 6.4% at $M = 40$. Which planner achieves the lower AoI depends on the geometry, yet the fleet size each would recommend is, in most regimes, the same.

A published competitor. The cluster patrol is the most different of these planners, and the only one taken from the literature. On AoI neither it nor the annealer dominates. At each planner’s own optimum the annealer’s median J is 32% lower on clustered fields (lower in 59 of 72 deployments), while the cluster patrol’s is 4% lower on the ring (54 of 72) and 31% lower on the core layout (55 of 72); at matched fleet sizes the cluster patrol’s J is 1.5% lower in median over all 2,844 runs. On the representative cell (paper, $M = 100$, 1.5 MJ) the annealer’s mean J is lower at every fleet size swept, by 33% at $K = 4$. The two win through different mechanisms, which the decomposition of Section VII-K separates (two instances per layout, $M = 100$, outside the evaluation seeds). On the core layout the cluster patrol’s fixed loops revisit every sensor at an exactly constant interval, losing nothing to irregular revisits, and its hops are shorter (18 against 27 s); the annealer loses 18–19% of its objective to irregular revisits. On the clustered layout the annealer wins through commute efficiency: sharing the whole field, most of its sorties stay near the depot’s dense area (commute radius 0.9–1.9 km against 4.0–4.7 km for fixed territories), which gives it 45% more visits per hour. Allocation by priority is not the lever in either case: with priorities spread over [1, 10], even the ideal square-root allocation is worth at most 7% over visiting every sensor equally often. Fixed territories buy regularity at the price of commute; field-wide

scheduling buys commute efficiency at the price of regularity. Its fleet-size optimum is less close to the annealer's — within one UAV in 72% of deployments ([68, 77]%), on average 1.8 UAVs smaller and 3.7 smaller on the core family, where the finite-window optimum is already known to depend on the planner (Section VII-C) — but closer to the rule: within one UAV of the point predictor in 84% ([80, 88]%), more often than the annealer's by 7.9 points (paired, [3.2, 12.5]). At $M = 40$ it is within one UAV of the rule in 23 of 24 instances. A published policy with a different structure, fixed territories instead of shared age-aware scheduling, therefore reaches an optimum the rule predicts at least as well as ours. The cluster patrol is the exception to a pattern the tour patrol shows clearly: the tour patrol and the annealer, whose J differs by up to 23%, agree more closely with each other about K^* than either does with the closed form, and the difference is significant as a paired comparison on the same deployments — the rate at which they lie within one UAV of each other exceeds the rate at which the annealer lies within one of the rule by 14.4 points (95% interval [11.1, 17.6]), and the tour patrol's by 10.6 points ([7.4, 13.9]). The cluster patrol instead agrees more often with the rule than with the annealer.

A planner built from the comparison. The diagnosis above suggests keeping fixed territories, which buy regularity, while choosing them to cut commute. We built such a planner, a depot-aware territorial scheduler (C4). Each UAV owns a disjoint sensor set and flies a fixed cyclic visit sequence, cut into energy-feasible sorties at launch by the same rule as the cluster patrol. The sets and tours are chosen before deployment by local search — relocating sensors between UAVs, re-optimising tours, assigning frequency classes — on a per-UAV simulation surrogate that uses only planner-side information: positions, priorities, system parameters and nominal generation rates. The search starts from the published k -means partition and from equal angular sectors and deploys whichever the surrogate prefers; without search, the first start reproduces our adaptation of the published method exactly. Unlike the published method, and like the annealer, it uses priorities. It was designed on separate pilot seeds with every setting fixed in advance, and evaluated once on the 216 held-out deployments against a criterion registered beforehand: a median own-optimum J below the cluster patrol's in *each* family, with own optima taken on the union of both planners' K values and extended until each planner's three extreme values exceed 1.5 times its minimum, so that no optimum is censored. A defect in the surrogate's clock surfaced during the first evaluation attempt; it was reproduced and fixed on pilot data, and the evaluation was rerun from scratch with no comparative statistic computed in between. The criterion holds in every family. Median own-optimum J is 21.0% lower on clustered fields ([18.2, 26.3]%, all 72 deployments), 10.6% lower on the core layout ([9.5, 12.1]%, all 72) and 6.6% lower on the ring ([4.4, 8.1]%, 60 of 72), 204 of 216 overall. The ring is the thin case: one ring cell ($M = 100$, 1.5 MJ) is a tie, and the largest single loss is 9.4%. At the rule's fleet size ℓ instead of each planner's own optimum the margins are 26.9, 10.6 and 3.4%, so the gain is not a by-product of choosing a different K . The partition alone does not carry it: equal

sectors without search lose to the published method on the core (+10.2%) and ring (+2.4%) layouts, so the gain comes from the surrogate-guided search, and frequency classes are nearly inert (a median 4% of sensors leave the base class). Two other designs failed: a selector choosing between planners by regime separated the synthetic families rather than the regimes and broke with an off-centre depot, and four hybrids (the annealer within territories, a square-root-due patrol, balanced territories, a quadratic index) were each worse than one of their parents. For fleet sizing, the stronger planner's optimum is predicted at least as well as the published one's: within one UAV of the point predictor in 88% of deployments (exactly in 54%), against 84% (49%) for the cluster patrol, and in 99% on clustered fields, 13.9 points above the cluster patrol there ([5.6, 22.2]). Against the annealer, in a secondary comparison registered with the criterion but not part of it, the result is mixed. C4's own-optimum J is lower by a median 39.6% on the core layout ([31.0, 45.8]%, 72 of 72 deployments) and 9.6% on the ring ([7.7, 11.7]%, 64 of 72), but not reliably on clustered fields: 5.3% in median with an interval that includes zero ([−10.3, +2.1]%, 40 of 72), and the annealer wins both clustered cells at 1.5 MJ outright, by 11.8% at $M = 100$ and 26.4% at $M = 200$, in all 12 deployments of each. C4 therefore does not supersede the annealer: field-wide, age-driven scheduling remains preferable on clustered fields under tight energy, the same split by mechanism as between the annealer and the published method. The two planners' optima lie within one UAV of each other in 84% of deployments. Both were measured under the same rule, and C4 is deterministic across machines (to one unit in the last place), so its runs are comparable with the annealer's, which all come from one machine.

Why the patrol wins on the ring. Writing the weighted age at the realised visit frequencies f_i and inter-visit variability CV_i as $J_2 = \sum_i w_i(1 + CV_i^2)/(2f_i)$ and factoring it as $J_0 E_{\text{alloc}} E_{\text{reg}}$, with $J_0 = (\sum_i \sqrt{w_i})^2/2\Gamma$ the ideal square-root allocation at the realised total rate, $E_{\text{alloc}} = \sum_i w_i/(2f_i)/J_0$ and $E_{\text{reg}} = J_2/\sum_i w_i/(2f_i)$, reproduces the simulator's measured age objective to within 0.5% on 15 of the 16 runs we decomposed and within 1.3% on the last — an independent numerical check of the visit-frequency model underlying Proposition 4, including the inspection-paradox correction; it does not test whether the realised frequencies are optimal. Applied to the ring ($M = 100$, 3 MJ, $K = 4, 6, 8$, two seeds), it attributes the patrol's advantage almost entirely to the visit rate Γ : the product of the three factors matches the measured ratio of objectives to within 4%, the rate factor tracks it in every instance, allocation favours the annealer by 1.4–4.3% as expected of a policy that ignores priorities, and regularity is secondary, ranging from a 2.8% loss to a 5.3% gain for the patrol and favouring it mainly at larger K . The rate difference comes from shorter hops: at $K = 4$ the patrol's inter-stop travel is 34.3 s against the annealer's 42.8 s, because consecutive stops on a tour are neighbours. We note the scope: this decomposition covers one cell and a two-run check at 1.5 MJ.

Better decisions do not monotonically improve the horizon. The CP-SAT planner makes the point directly. Tripling its

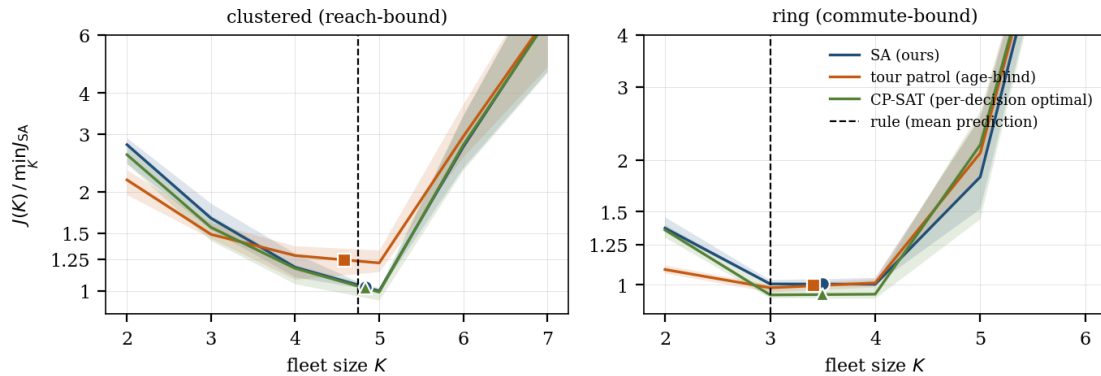


Fig. 4: Objective against fleet size under three planners, $M = 40$, $E_{\max} = 1.5$ MJ: median over 12 instances of $J(K)$ divided by the annealer’s minimum on the same instance, with interquartile bands. Normalising by one planner’s minimum keeps the level differences visible. Markers: mean over instances of each planner’s own optimum; dashed line: mean point prediction of the rule. The curves sit at different heights but reach their minimum at the same fleet size. On the ring the objective is nearly flat between $K = 3$ and 4 under every planner.

search budget raises the share of proven-optimal decisions on the ring from 0.81–0.90 to 0.92–0.99 at $K = 2$ –4, but changes J by between -5.4% and $+6.6\%$ with a median of zero; tripling again, which proves essentially every decision optimal, changes three of six runs by $+1.1\%$, $+2.4\%$ and -1.7% and leaves the other three bit-identical. A per-decision solution that is better, or even exact, changes the state the next decision starts from, and the rolling-horizon objective is myopic. Per-decision optimality is therefore not horizon optimality, and no per-decision solver, however exact, is a ground-truth oracle for J .

Resolution. That sensitivity sets how finely the ring optimum can be resolved. At $M = 40$, a fleet one UAV off the optimum costs a median 2.8% in J on the ring against 18% on clustered fields, so a perturbation of a few percent in J can move the argmin by one. At CP-SAT’s default budget the ring argmin was biased upward for exactly this reason (exact agreement with the annealer 0.50); at a threefold budget it rises to 0.67, and a further threefold budget leaves the argmin unchanged on every instance checked, so the reported ring figures use the converged budget. Even then one instance differs from the annealer by one UAV with every decision proven optimal: an exact myopic planner simply prefers the neighbouring fleet size there, by 1.2% in J . On commute-bound fields one-UAV resolution is therefore both what the measurement supports and all an operator needs.

Where the planners disagree. The finite-window core regime is the exception, and it is instructive. At core $M = 200$, 3 MJ, the patrol’s optimum is 27.67 against the annealer’s 28.25, both far above the rule’s 10.33: the stranding mechanism of Proposition 6 is a property of the problem. At 1.5 MJ the patrol places it at 10.75 against 13.67, within one UAV in only one instance of twelve. Once abandoning a peripheral sensor pays, the optimum balances that sensor’s age against the service the rest of the field receives, and the second term depends on how well the planner serves the shrinking reachable set. The mechanism transfers; its precise location does not.

TABLE XII: Kernel belief (M1) versus prior only (M2), paired over 12 instances, paper, $M = 100$, $E_{\max} = 1.5$ MJ, $K = 4$, per-leg SA.

$\bar{\tau}$	q	ΔJ	ΔJ_{age}	ΔJ_{event}	Δcatch	sign test
67 min	0	-1.2%	$+1.3\%$	-6.2%	$+2.4$ pt	n.s.
67 min	0.5	-0.4%	$+1.6\%$	-4.5%	$+1.7$ pt	n.s.
67 min	1	-1.3%	$+0.8\%$	-5.6%	$+3.3$ pt	n.s.
15 min	1	$+3.4\%$	$+5.4\%$	-5.2%	$+8.2$ pt	worse, $p = .006$
5 min	1	$+2.9\%$	$+3.2\%$	-4.7%	$+11.6$ pt	worse, $p < .001$

L. Value of event-state information

The reach bracket holds for any routing policy; the commute branch enters through the rate curve the policy induces. We measure the value of one additional information channel a policy can exploit beyond the age vector: explicit knowledge of where events are live. On paired instances at the rule’s $K = 4$ we compare the prior-only planner (M2, event-blind index) with the kernel-belief planner (M1), across event correlation q and event lifetime $\bar{\tau}$ (Table XII).

Three things follow. The gain from belief does not depend on q : spatial correlation carries no exploitable information at these parameters, and the $\approx 1\%$ at long lifetimes is the self-observation term (a visit clears the event), available to any planner. For events comparable in duration to a sortie, belief is harmful: the planner catches more events but sacrifices freshness, and the event term it protects is only 4–18% of J because most short events expire before any UAV can reach them. And K^* is unchanged under correlated events with the kernel planner (12/12 agreement).

What this does and does not bound is worth stating precisely. It bounds *one* channel of possible advantage—explicit knowledge of event state—within the information structure and event model of Section III: at these parameters that channel is worth $\leq 1\%$ and is negative for short-lived events. It does not bound advantages available from better sequencing, better sensor-to-UAV partitioning, or non-myopic energy use across sorties. The per-decision certificates of Section VI-A

Algorithm 4: Pre-deployment fleet sizing

Input: sensor positions x_i , weights w_i , nominal rates λ_i ; airframe v, P_f, P_h, R_u, B ; inventory U ; window T_b, T ; longest interruption to tolerate G

Output: a fleet size, or an interval to sweep

- 1 $r_i \leftarrow \|x_i - x_0\|$, $r_{\max} \leftarrow \max_i r_i$
- 2 $K_{\text{reach}} \leftarrow vU/(2P_f r_{\max} + vP_h B/R_u)$,
 $K_{\text{flight}} \leftarrow vU/(2P_f r_{\max})$ // reach bracket,
 (3)
- 3 solve $K = K_{\text{commute}}(\hat{r}_c(K), \hat{P}(K))$: $\hat{r}_c$ from the
 radial order statistic, $\hat{P}$ from hover conservation
 // Sec. VII-J
- 4 $K_0 \leftarrow \lfloor K_{\text{reach}} \rfloor$, $\ell \leftarrow \min(K_0, \text{round } \hat{K}_{\text{commute}})$
- 5 **if** $\ell < K_0$ **then return** ℓ // commute-bound
- 6 **if** (5) **holds then return** K_0 // reach-bound
- 7 **return** the interval $[K_0, K_{\max}]$ and sweep it, extending
 the sweep if the best point lies at its upper end
 // finite-window stranding may pay

and the CP-SAT comparison of Section VII-K close part of that remaining gap, but they reach $M = 40$, not the headline configuration, and set-orienting hardness grows quickly with M ; we therefore do not claim the planner is near-optimal at $M = 400$. The defensible statement is that explicit event-state information adds little over an age-based index at the fleet-size optimum, not that a learned policy cannot help.

VIII. DISCUSSION AND LIMITATIONS

Using the rule. Algorithm 4 collects the pre-deployment procedure. Every input is known before the fleet flies; nothing in it requires a simulation, a pilot sortie or a fleet sweep unless the design-time test flags the field. It uses the robust reach cap throughout, and when the test flags a field it returns the whole range $[K_0, K_{\max}]$ of (5) rather than Result 1's $[K_0, K_R]$, because K_R is the realised commute optimum and is not known before deployment; the flight-only alternative of Section VII-B is an operational choice that depends on the interruptions a deployment must tolerate, and is deliberately left out of the procedure.

Duty cycling. Proposition 1 shows the rule sizes the airborne fleet $K\delta$ rather than the airframe count, so an operator cannot evade the reach cap by buying more UAVs and flying them less. The corollary is that the two are interchangeable for freshness, and airframes beyond K^* are only useful as redundancy.

Scope of the energy model. The shared-budget assumption of Remark 1 is what makes fleet size non-monotone, and it is an assumption about a shared energy *stock*; a fixed charging *rate* bounds the airborne count by a different mechanism that we do not model; under independent per-airframe batteries with no other shared bottleneck neither cap yields a finite optimum, since the reach radius no longer shrinks with K and the fleet visit rate is monotone increasing (Remark 1). Equal allocation (Remark 2) is likewise a modelling choice rather

than an optimum; reserving one sortie for the peripheral sensor removes the stranding regime in the two core cells tested (Sec. VII-C), and the general allocation problem remains open.

Modelling. Results rest on a single simulator and on our placement and energy models. The reachability bracket is geometric and transfers to any field shape and depot location through $r_{\max}$, but the full rule does so only where a field-average commute radius represents the field: with an off-centre depot the ring layout's far side binds through commute cost and the rule is one UAV too high (Sec. VII-F); the commute cap and the boundary transfer with the realised $(r_c, \bar{P})$ but the closed form's constant-cost assumption does not, as the 3 MJ ring cells show. Wind and altitude changes are not modelled; depot position is tested in Section VII-F. Validation on a real sensor layout is the natural next step.

Objective. Age is reset at collection, not delivery. Deconfliction near the depot is not modelled; earlier work by the authors measured its effect at 0–7% of a static-mission objective, and asynchronous relaunch reduces near-depot congestion, but at large K (the 6 MJ cells) it may matter.

Planner dependence. The reach bracket holds for any routing policy; which point inside it a planner sustains does not. The commute branch enters through the rate curve a policy induces, and Section VII-K measures how much that matters: little for the location of the optimum in most regimes, a great deal for the level of J , and decisively in the finite-window core regime, where planners agree that stranding pays but not on how far. On commute-bound fields the objective is flat enough near the optimum (2.8% per UAV) that no planner we tested resolves it more finely than one UAV at $M = 40$.

Statistics. The per-instance statistic is what a step-function rule requires; population means blur it. Agreement rates and regret are the primary evidence, and their 95% intervals come from a bootstrap that resamples instances within each cell, keeping the cell design fixed (10,000 replicates; exact Clopper–Pearson intervals where a rate is 0 or 1). Per-cell intervals, with 12 instances each, are wide — six agreements in twelve spans roughly [21, 79]% — and are given in the supplementary data rather than in Table IV; the per-cell sign tests are descriptive, reported per cell without correction for multiplicity, and the lower-side test is at $p < .001$ in 19 of the 20 primary cells. The two exceptions are explained by mechanisms rather than noise.

IX. CONCLUSION

The number of UAVs a continuously operating data-collection fleet should keep airborne is bounded by two physically distinct mechanisms, and the boundary $r_{\max}/r_c = 1 + \sqrt{\bar{P}/P_f}$ predicts which one binds. The reach cap is a feasibility statement about the farthest sensor, exact as a bracket between its full-buffer and flight-only limits, and duty cycling relocates it to the airborne count rather than removing it (Proposition 1). The commute branch rests on an exact allocation principle — cost-weighted square-root frequencies, under which minimising weighted age is equivalent to maximising the fleet's visit rate, which under a fixed normalised cost profile is proportional to productive energy per unit time

divided by the mean energy a visit costs (Proposition 4) — together with a first-order closed form that approximates it. We are explicit about which is which: in the layout family where the closed form misses by two UAVs, the principle holds exactly, and the error is traceable to a single measurable quantity, the K -dependent inter-stop travel time that Proposition 3 isolates from the dwell time it proves irrelevant.

The rule was validated per instance across three layout families, four sensor counts, three budgets and five field sizes. Its failure modes were characterised rather than averaged over: a stranding condition prices when exceeding the coverage bound is worthwhile, and a pre-deployment criterion, computable from the site survey and known system parameters, flags in advance the deployments where the reach branch may not govern within the operating window. Fleet coordination through shared commitments is a prerequisite for the measurement rather than a refinement, costing a factor of 2.6 in the objective and a whole UAV in the optimum when omitted. Finally, explicit event-state information adds little over an age-based index at the optimum within the event model studied. The optimum itself largely survives a change of planner: an age-blind patrol and a planner that proves almost all its decisions optimal locate it within one UAV of ours in most deployments while achieving materially different AoI, and the rule predicts their optima about as well as it predicts ours. The exception is instructive. Where stranding a peripheral sensor pays, every planner we tested places the optimum far above the reach cap, but the precise point is planner-dependent; the mechanism is structural, its location is not.

APPENDIX A

DERIVATION OF THE COMMUTE CAP

With $a = P_f t_c$ and $b = t_c(\bar{P} - P_f)$, (6) is $\Gamma \propto f(K) = K(U - aK)/(U + bK)$. Then $f'(K) \propto (U - 2aK)(U + bK) - b(UK - aK^2) = U^2 - 2aUK - abK^2$, whose positive root is $K = U(\sqrt{a^2 + ab} - a)/(ab) = \frac{U}{a} \cdot \frac{\sqrt{1+b/a}-1}{b/a}$. Since $b/a = \bar{P}/P_f - 1$, $\sqrt{1+b/a} = \sqrt{\bar{P}/P_f}$ and $K = \frac{U}{a(1+\sqrt{\bar{P}/P_f})} = \frac{U}{t_c(P_f + \sqrt{P_f \bar{P}})}$, which is (7). With sublinear yield $n = c(E_u - a)^\alpha$ the maximiser of $Kn/(t_c + n\tau_s)$ has no closed form; it is computed numerically from measured (c, α, τ_s) , and in the fitted ring cells of Table XIII it lies above (7); we have not shown that this holds for $\alpha < 1$ in general.

APPENDIX B

SUPPLEMENTARY DIAGNOSTICS

Choice of estimator. For transparency, the alternatives and what they would report: taking the argmin of mean J per cell gives exact agreement 0.504 and within-one 0.735; of median J , 0.500 and 0.812; of mean relative regret $J(K)/J(K_{\text{inst}}^*)$, 0.489 and 0.735; against 0.581 and 0.820 for the per-instance estimator we adopt. The four disagree on the summarised optimum in 9 of 24 cells. Note that the robust alternatives (median, relative regret) do not outperform the plain mean, so the gap is not an outlier effect; it is the granularity mismatch above. A practitioner sizing a fleet for one deployment faces the per-instance question; an operator choosing a single K

TABLE XIII: Diagnosing the commute branch: measured optimum against the closed form ($\alpha = 1$) and against a fitted sublinear stop count. Ring layout.

cell	measured	$\alpha = 1$	α -corrected	fitted α
$M=50, 3 \text{ MJ}$	8.50	6	8	0.68
$M=100, 3 \text{ MJ}$	7.75	6	8	0.71
$M=200, 3 \text{ MJ}$	6.92	6	7	0.85
$M=100, 1.5 \text{ MJ}$	3.08	3	4	0.87

for a portfolio of future deployments faces the aggregate one, and should then also compare against a rule evaluated at the population geometry rather than at each instance's.

A fitted commute exponent. Fitting $n = c(E_u - P_f t_c)^\alpha$ on the stranding-free rows and re-maximising recovers the missing UAVs (Table XIII): the corrected form recovers the whole ring family at 3 MJ, the three cells where the rule misses worst, while the uncorrected $\alpha = 1$ form remains the better predictor at 1.5 MJ. We report this as a diagnosis rather than adopting it as the rule, for a reason that is worth stating plainly: α is *fitted*, and a rule whose claim is that it contains no fitted constants should not acquire one to recover three cells. Deriving α from the layout geometry — as Proposition 5 derives η — would make the commute branch both parameter-free and accurate, and is the natural next piece of theory. We tested the most promising candidate derivation and report it as a negative result, because the way it fails is informative. If the planner's marginal insertion costs X have a locally d -dimensional law $F_X(x) \sim Ax^d$, then cost-aware selection at a value-per-energy threshold gives $n \propto \theta^{-d}$ and $E_{\text{hop}} \propto \theta^{-(d+1)}$, hence $E_{\text{hop}} \propto n^{(d+1)/d}$ and $\alpha_{\text{geom}} = d/(d+1) = 2/3$ for a locally planar layout — strikingly close to the 0.68 and 0.71 observed. Per-sortie regression refutes it: $\log E_{\text{hop}}$ on $\log n$ over stranding-free sorties gives slopes 0.49, 0.98 and 0.82 (95% intervals [0.43, 0.55], [0.88, 1.07], [0.76, 0.89]), all excluding the predicted 1.5 and all below unity: marginal stops get *cheaper* in hop energy, not dearer, which is the opposite sign to the mechanism.

The decomposition into hop and hover energy then locates the difficulty, and it is not where the fitted exponent suggests. Hover energy per sortie does grow superlinearly in n ($\beta_{\text{hov}} = 1.16, 1.09, 1.03$), consistent with the conservation argument of Proposition 3: a fixed total hovering requirement H shared over a fleet visit rate $\Gamma(K)$ gives $E_{\text{hov}}/n = P_h H/\Gamma(K)$, which we verify to a constant factor of 1.32 ± 0.01 across the three cells — the proportionality holds, the constant does not. But hover is only 6–23% of productive energy here, too small a share to produce the aggregate exponent. Decisively, *measured* per-sortie productive energy is *sublinear* in n ($\beta_p = 0.55, 0.95, 0.86$) while the aggregate fit implies $1/\alpha = 1.41, 1.47, 1.18$. The two disagree in direction. The resolution is that $E_u - P_f t_c$ is a budget proxy, not energy actually spent: sorties do not exhaust their allowance and t_c is a cell-level mean. We therefore conclude that α is an elasticity of the closed loop — routing geometry, visit-rate feedback and backlog state compressed into one number — rather than a geometric constant of the layout, and we do not attempt to derive it.

APPENDIX C REPRODUCIBILITY

Every number in this paper is traceable to a row in the released CSV files keyed by (layout, M , $E_{\max}$, K , seed, coordination, replanning, q , belief, $\bar{\tau}$, window) and to the commit hash of the simulator. A single command, `python reproduce.py`, regenerates every table, statistic, interval and figure from those files without running the simulator; the bootstrap uses a fixed seed, so the outputs are identical run to run. Code: [\[repository URL\]](#).

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